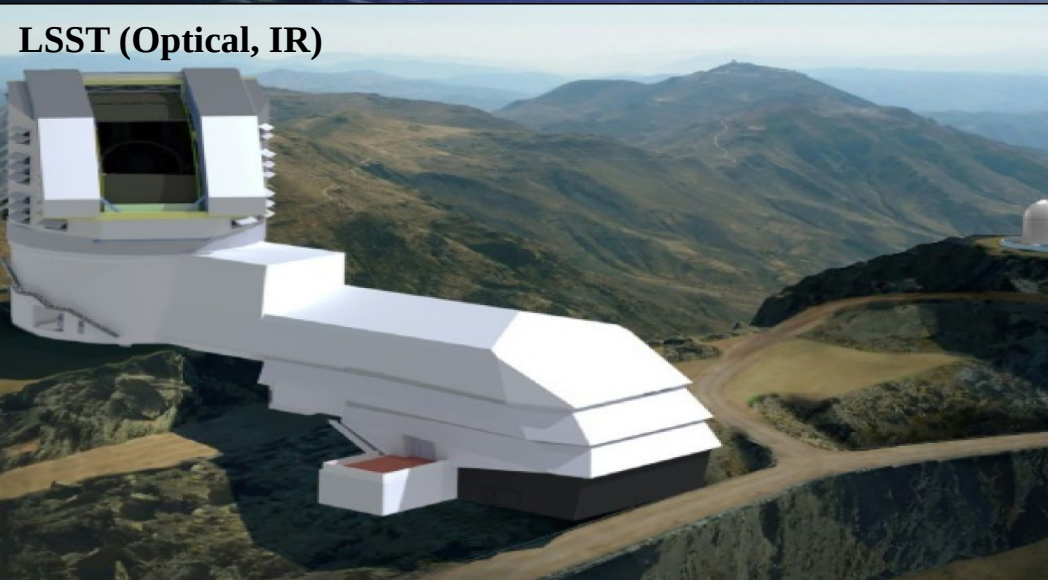
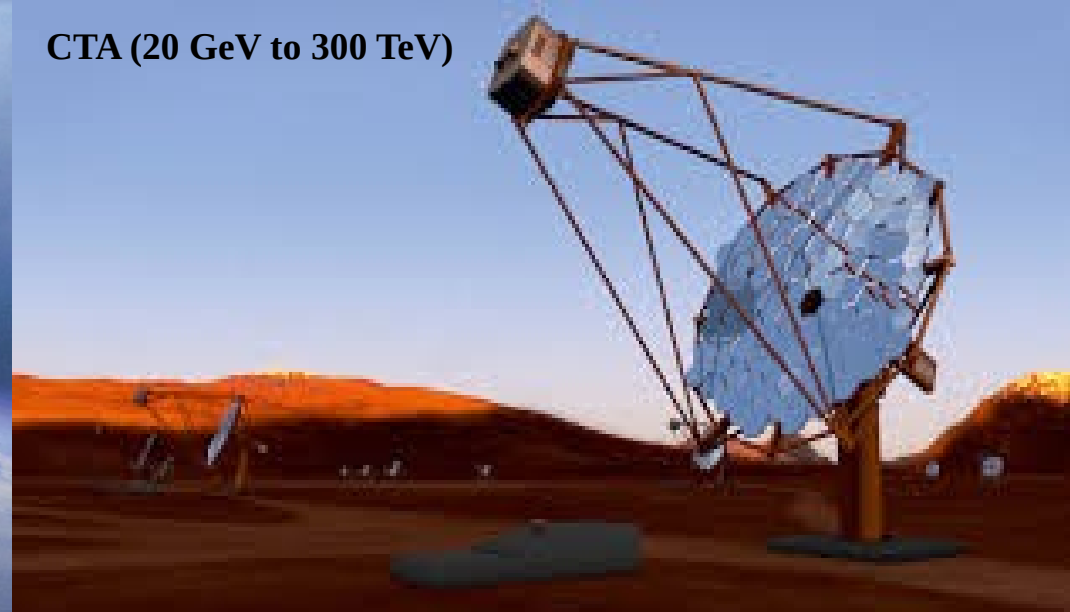
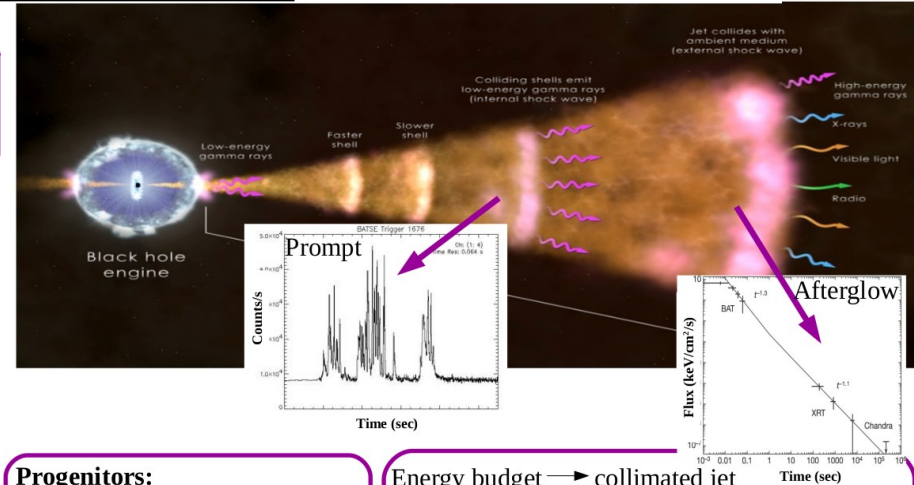




Study of Gamma-ray bursts/ Fermi

Study the spectral and temporal properties of the prompt emission phase of GRBs observed by Fermi

- Data analysis
- Consideration of systematic and statistical effects
- likelihood maximization



Novelties !

- Increase of the break energy from MeV to GeV
- Proof of an internal origin of the GeV emission observed during the keV-MeV prompt phase
- A&A article: [2017AA...606A..93Y](#)

Collaborators: F. Piron, R. Mochkovitch, and F. Daigne

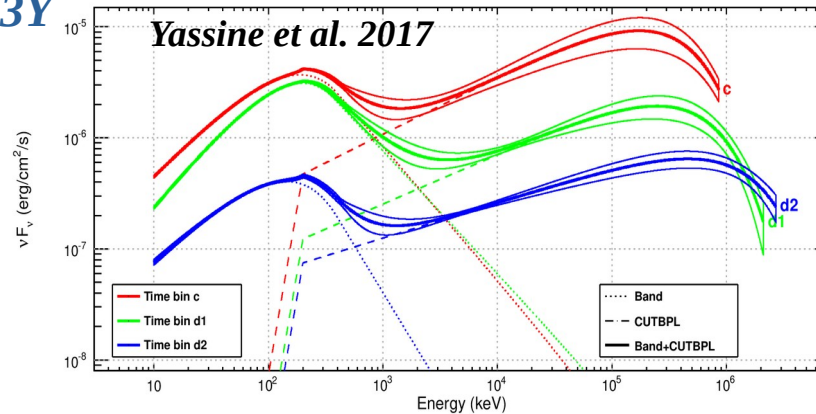
Progenitors:

LGRBs ($> 2s$): massive stars
SGRBs ($< 2s$): NS/NS or NS/BH mergers

Energy budget → collimated jet

Jet energy dissipation (e.g. internal shocks): prompt phase ($R \sim 10^{14-15}$ cm)

Jet interaction with the ambient medium (external shocks): afterglow phase ($R \sim 10^{16-17}$ cm)



Study of Gamma-ray bursts/ Fermi

Interpretation of the GRB observed properties using an internal shock model

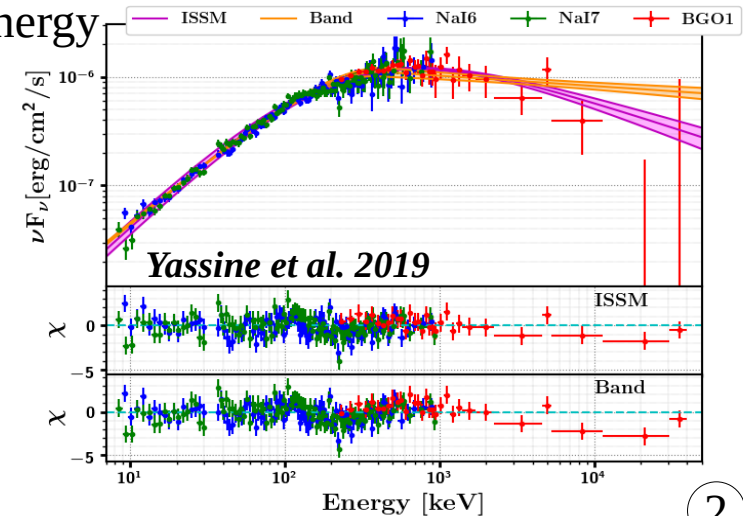
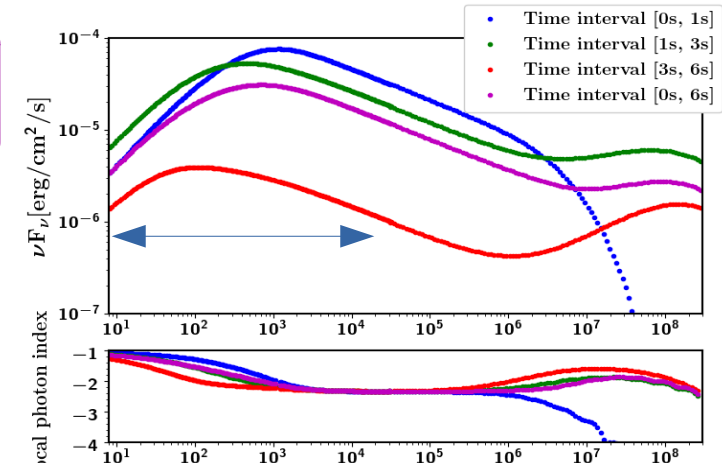
- Simulate the Fermi response to a synthetic burst (IAP IS model)
- Find a parametrization of the simulated spectra
- Apply the parametrization function to real data (74 GRBs)

Novelties !

- Fit the local photon index of the synthetic spectra as a function of energy
- Build a representative function (ISSM) of the IAP model spectra:

$$\frac{dN_{\text{ISSM}}}{dE}(E) = \frac{A_{\text{ISSM}}}{\left[1 - \frac{E_p}{E_r} \left(\frac{2+\beta}{2+\alpha}\right)\right]^{\beta-\alpha}} \times \left(\frac{E}{E_r}\right)^{\alpha} \left[\frac{E}{E_r} - \frac{E_p}{E_r} \left(\frac{2+\beta}{2+\alpha}\right)\right]^{\beta-\alpha}$$

- The **ISSM** model is a good model for 81% of the GRBs
- A&A article: [640, A91 \(2020\)](#)



Study of Gamma-ray bursts/ Fermi & CTA

Estimation of the detectability of GRBs by the Very High-Energy instruments.

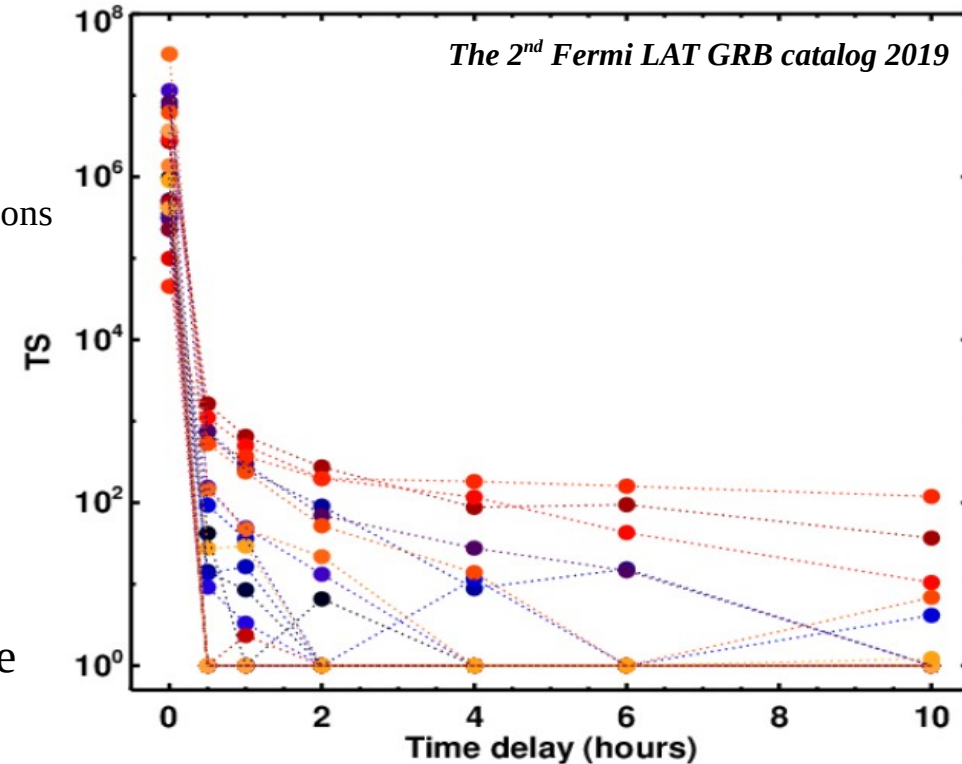
- Study 33 long GRBs with redshift measurements
- Include $\gamma\gamma$ absorption effect (annihilation of high-energy photons with photons from the ExtraGalactic Background Light)

Results !

- The CTA telescopes would readily detect GRBs given ideal observational conditions during night time

Additional contributions !

- ★ Contribution to the Burst Advocate monitoring service (BA)
- ★ Contribution to the Data Quality Monitoring shifts



Vera C. Rubin Observatory (LSST/DESC)



- 8.4m telescope 9.6 degree² field-of-view
- Survey the south hemisphere of the sky every 3-4 nights

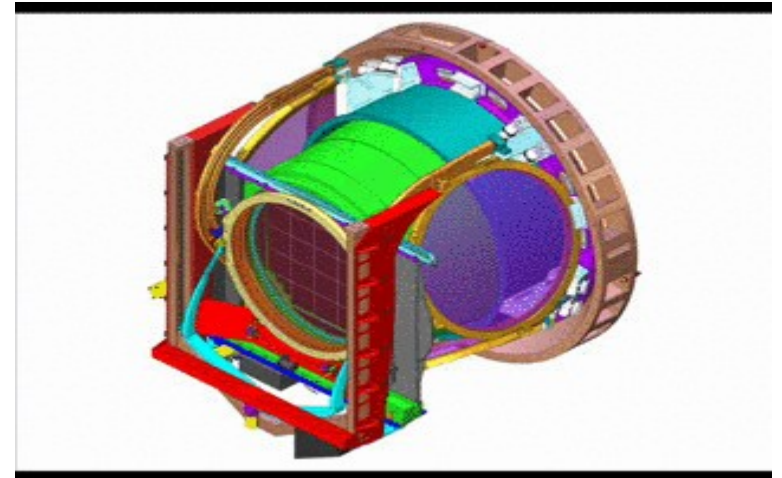
The LSST Surveys :

Wide-fast-deep (WFD)

- 18,000 sq deg
 - ➔ A given object is observed in a given filter once every three days
 - ➔ Each field will be seen twice a night (2 visits) separated by 15-20 mins

Several Deep Drilling Fields (DDF)

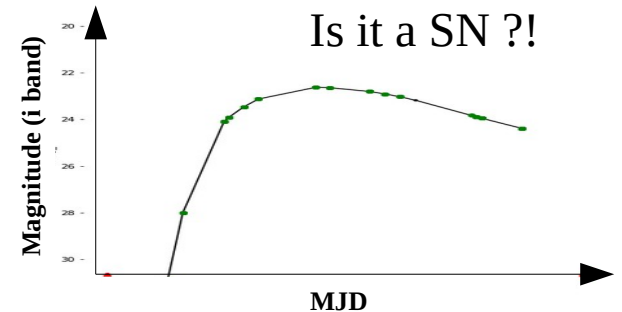
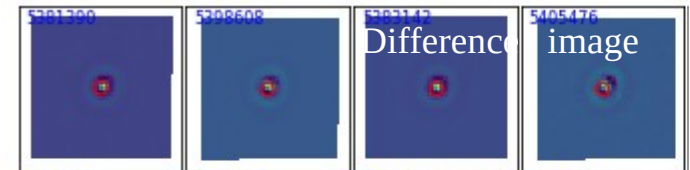
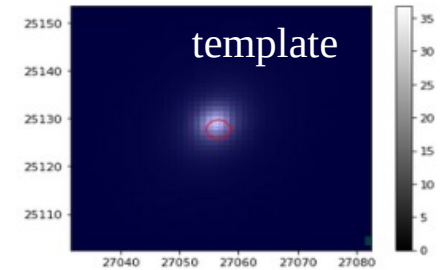
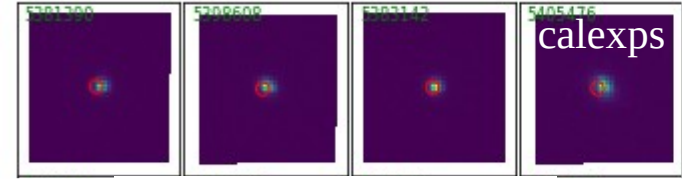
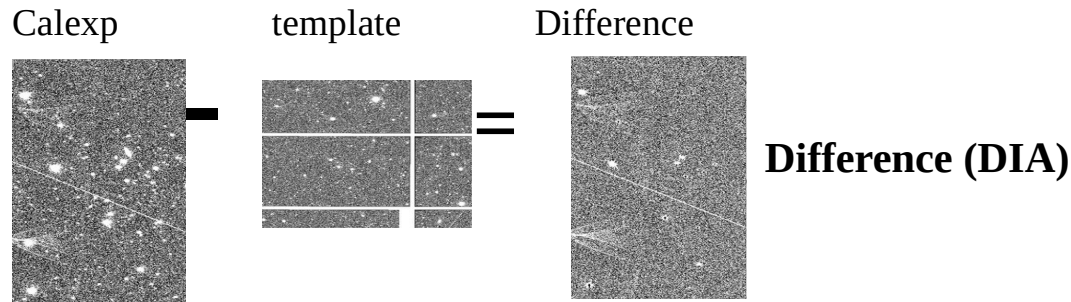
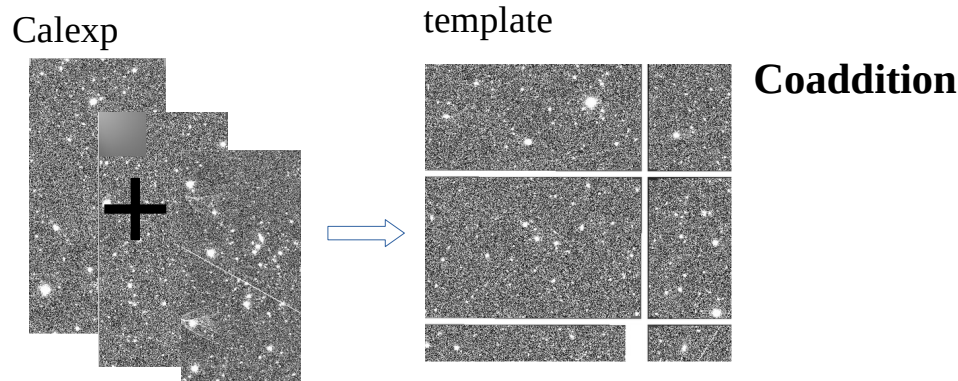
- Each of 10 sq deg
 - ➔ A given object is observed in a given filter several times during one night



Validation of the SN detection pipeline (simulated data)

Compare the truth SN LCs with the measured ones

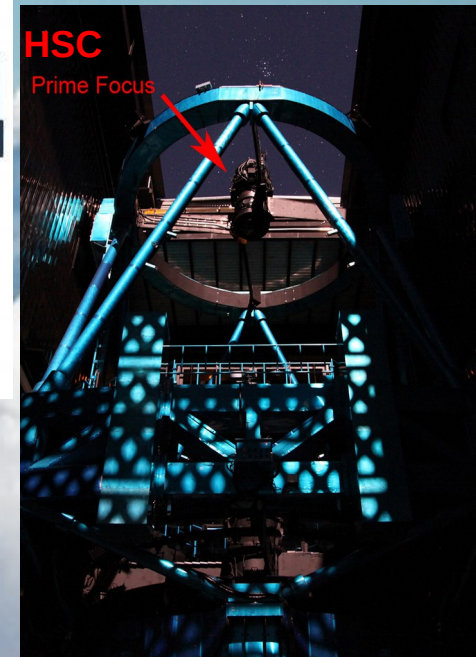
- Match between the truth objects with diasrc and diaobject catalogs
- Match to coadd, calexp and difference images
- Look at the flags of the diasrc in case of bogus detection



Validation of the SN detection pipeline (real data)

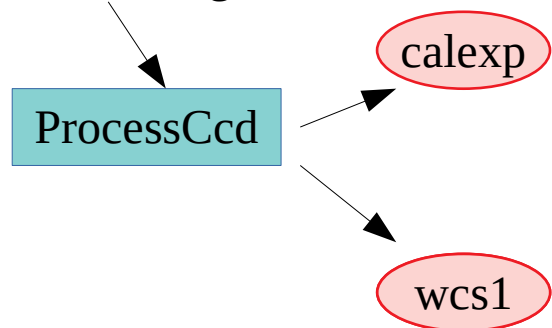
- Produce calibrated light curves from the HSC transient survey data
- Pipeline: LSST pipeline v20.0.0
- Data: COSMOS UDEEP field
- (PDR1, filters Z & R)

Subaru Telescope (8.2 m)



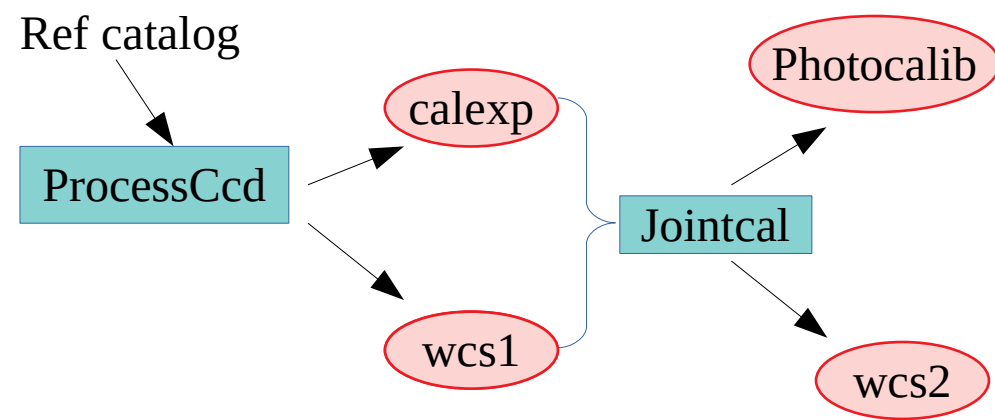
Pipeline

Ref catalog



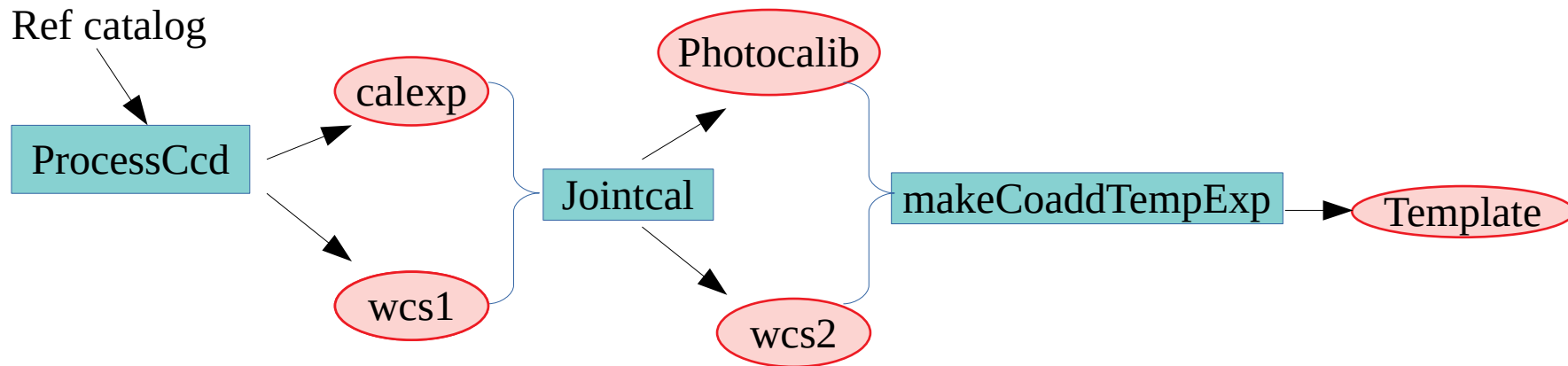
- Reference catalog for ProcessCcd: Pan-STARRS (ps1_pv3_3pi_20170110)
- Jointcal ran night by night (1 year from each other, proper motion is not handled)
- Reference catalog for Jointcal: Pan-STARRS (Gaia recently included)
- Coaddition templates were done night by night

Pipeline



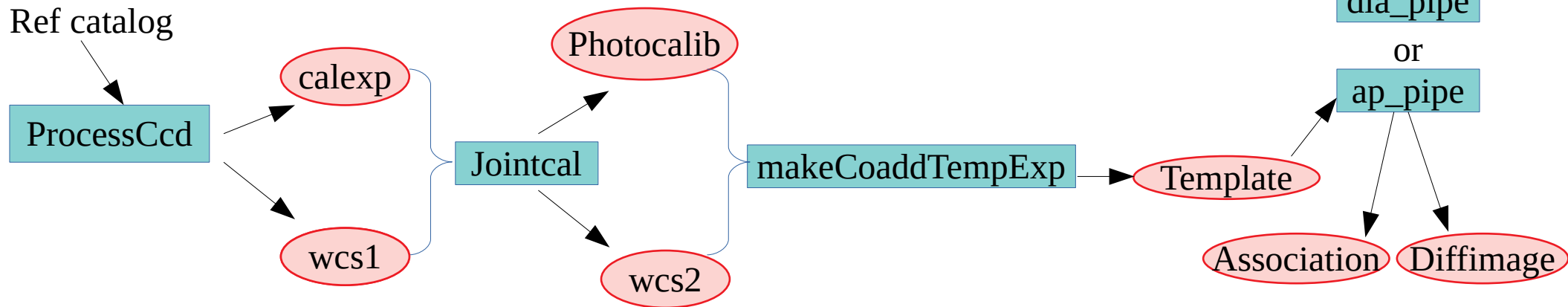
- Reference catalog for ProcessCcd: Pan-STARRS (ps1_pv3_3pi_20170110)
- Jointcal ran night by night (1 year from each other, proper motion is not handled)
- Reference catalog for Jointcal: Pan-STARRS (Gaia recently included)
- Coaddition templates were done night by night

Pipeline



- Reference catalog for ProcessCcd: Pan-STARRS (ps1_pv3_3pi_20170110)
- Jointcal ran night by night (1 year from each other, proper motion is not handled)
- Reference catalog for Jointcal: Pan-STARRS (Gaia recently included)
- Coaddition templates were done night by night

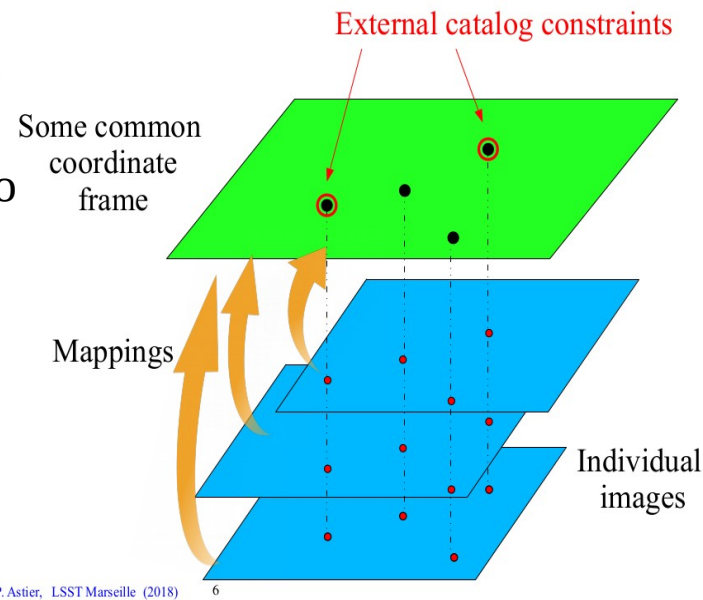
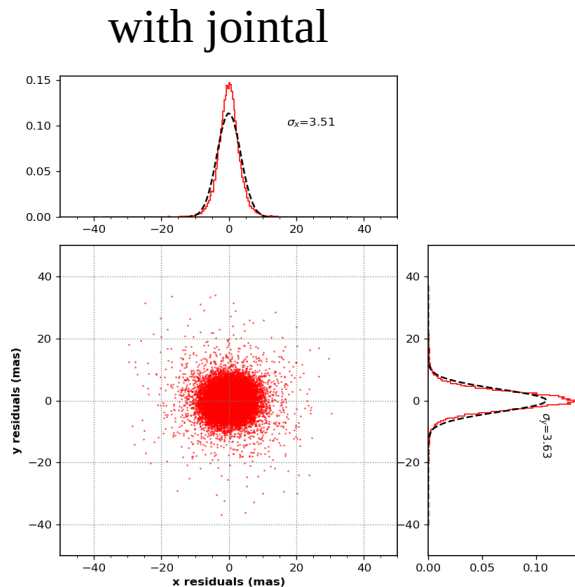
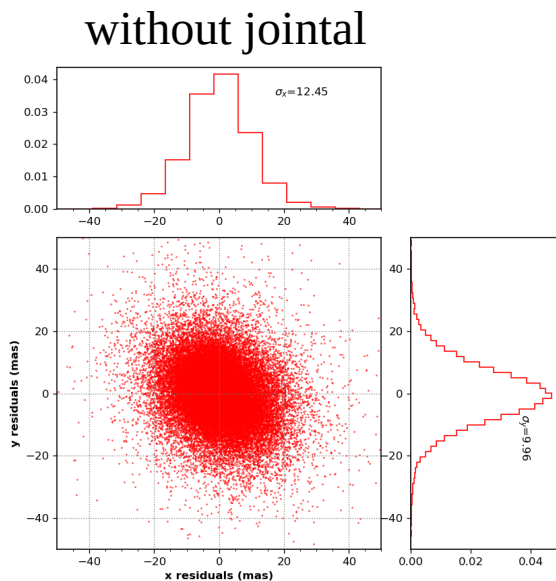
Pipeline



- Reference catalog for ProcessCcd: Pan-STARRS (ps1_pv3_3pi_20170110)
- Jointcal ran night by night (1 year from each other, proper motion is not handled)
- Reference catalog for Jointcal: Pan-STARRS (Gaia recently included)
- Coaddition templates were done night by night

Jointcal: astrometric results

- HSC is challenging for combinatorial astrometric matching (coaddition), because of huge optical distortions.
- Jointcal: Simultaneous optimization of Astrometry & Photometry for thousands of Exposures
- Mappings to some undistorted space (e.g. tangent plane) allows one to remove the effects of optical distortions.



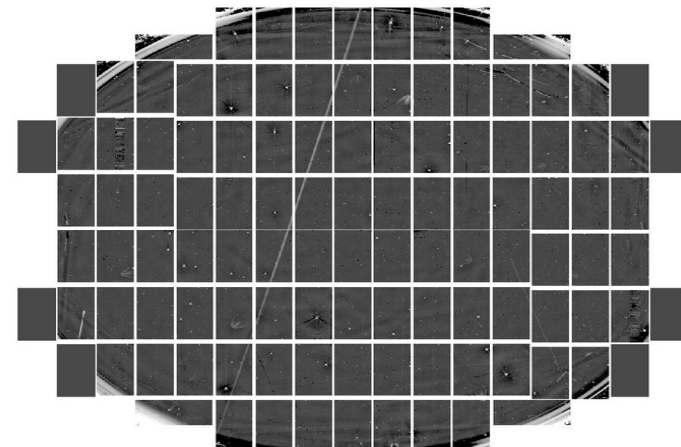
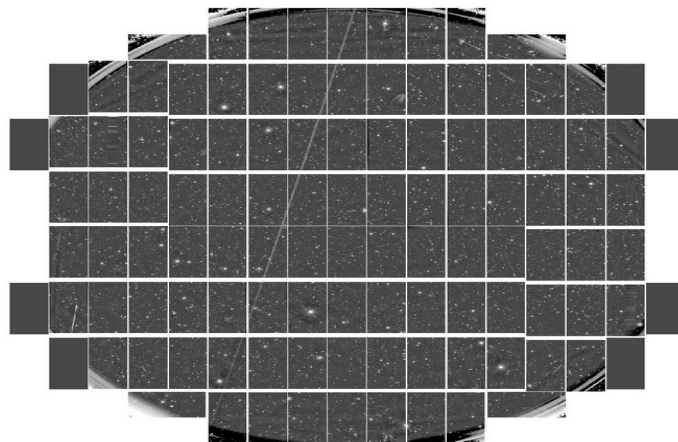
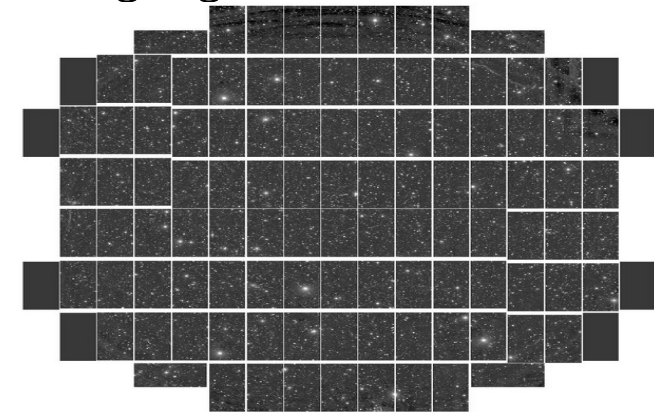
Jointcal reduces the error distribution by ~3 orders of magnitude

Apply the method on a larger HSC dataset

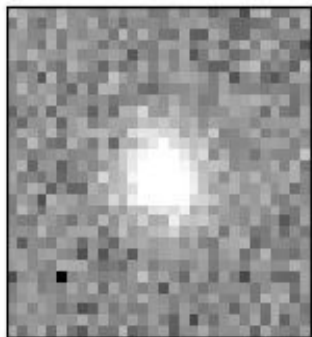
build templates from good seeing nights

Subtract higher cadence nights

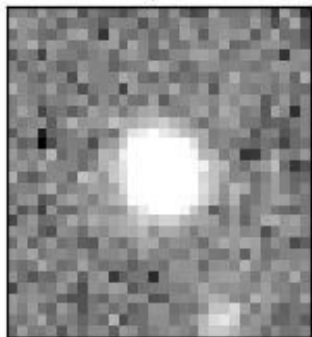
Get difference image



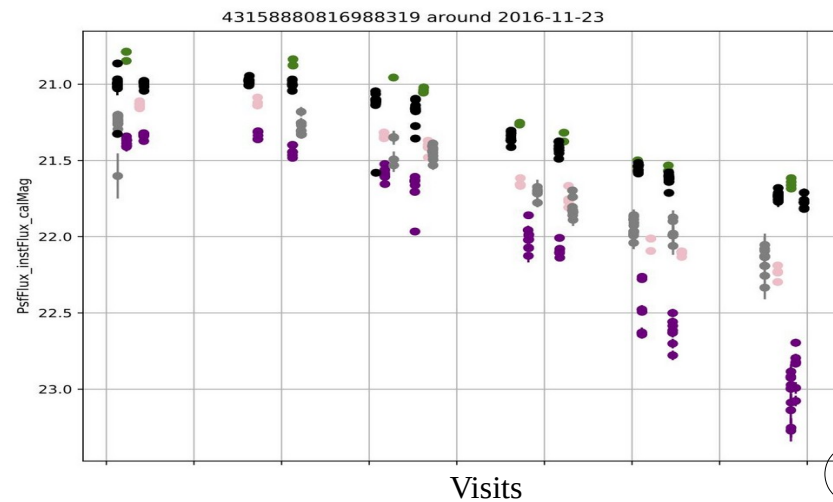
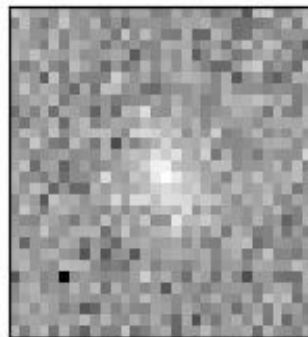
Processed



Template



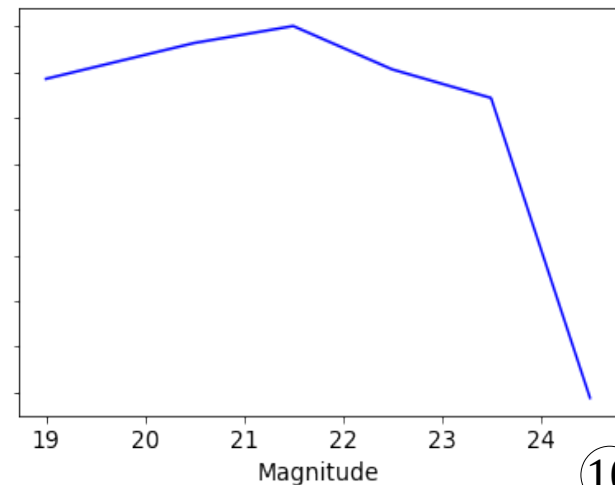
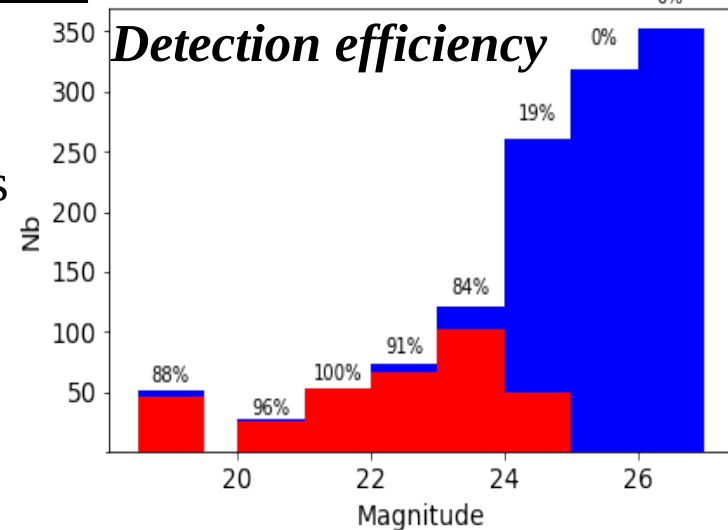
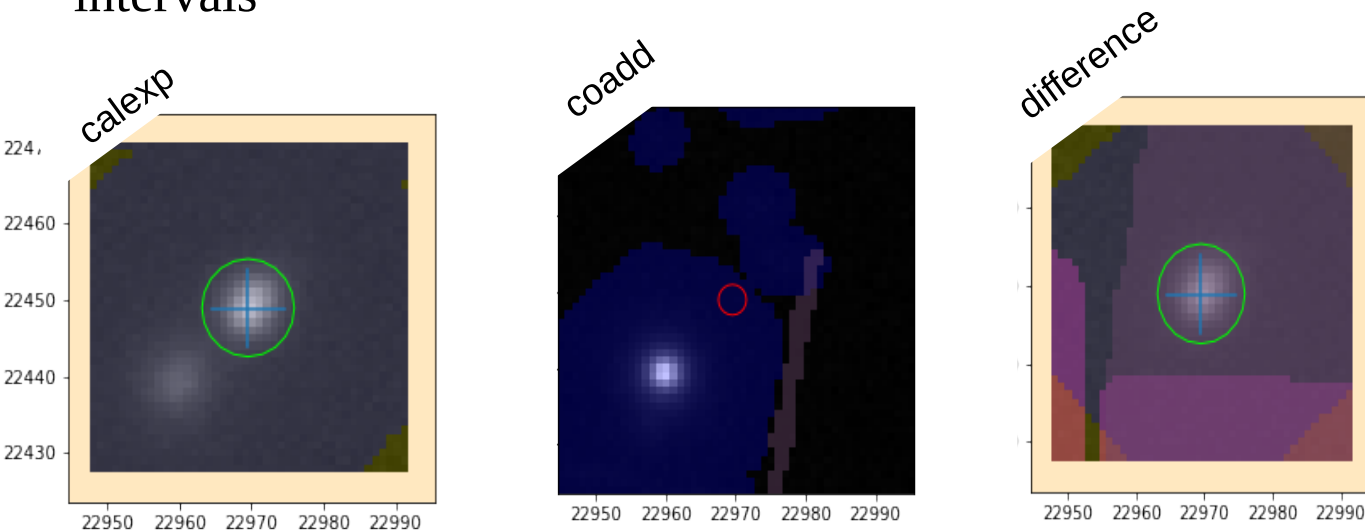
Difference



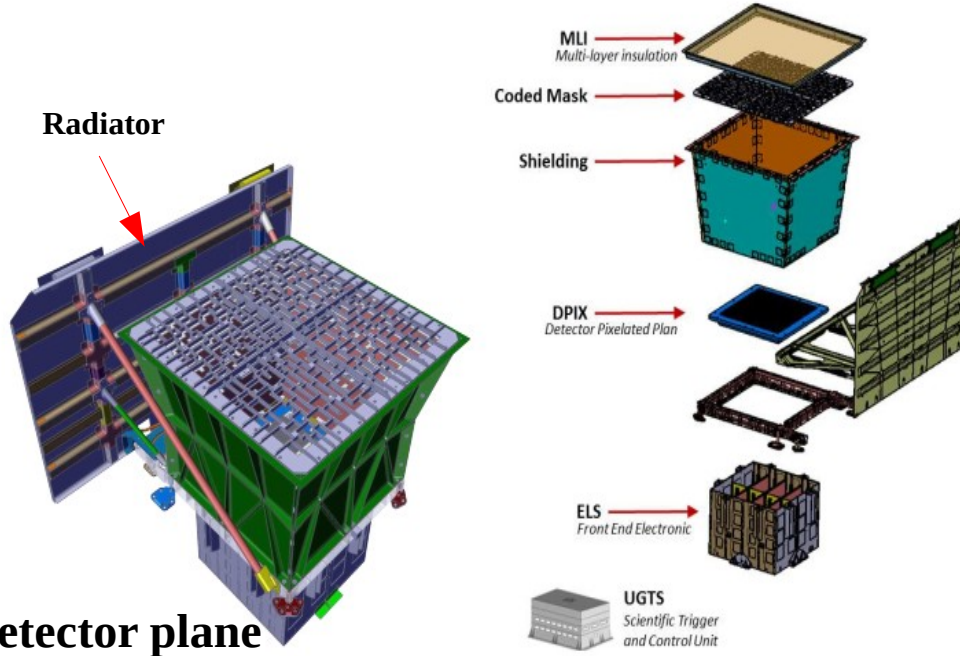
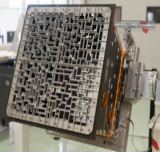
[Explore here](#)

Validation of the DIA pipeline: Fake injection

- Inject fakes on to science images (in calexp)
- Subtract images from a coadd templates of stacked exposures
- Check the efficiency of detection in different magnitude intervals

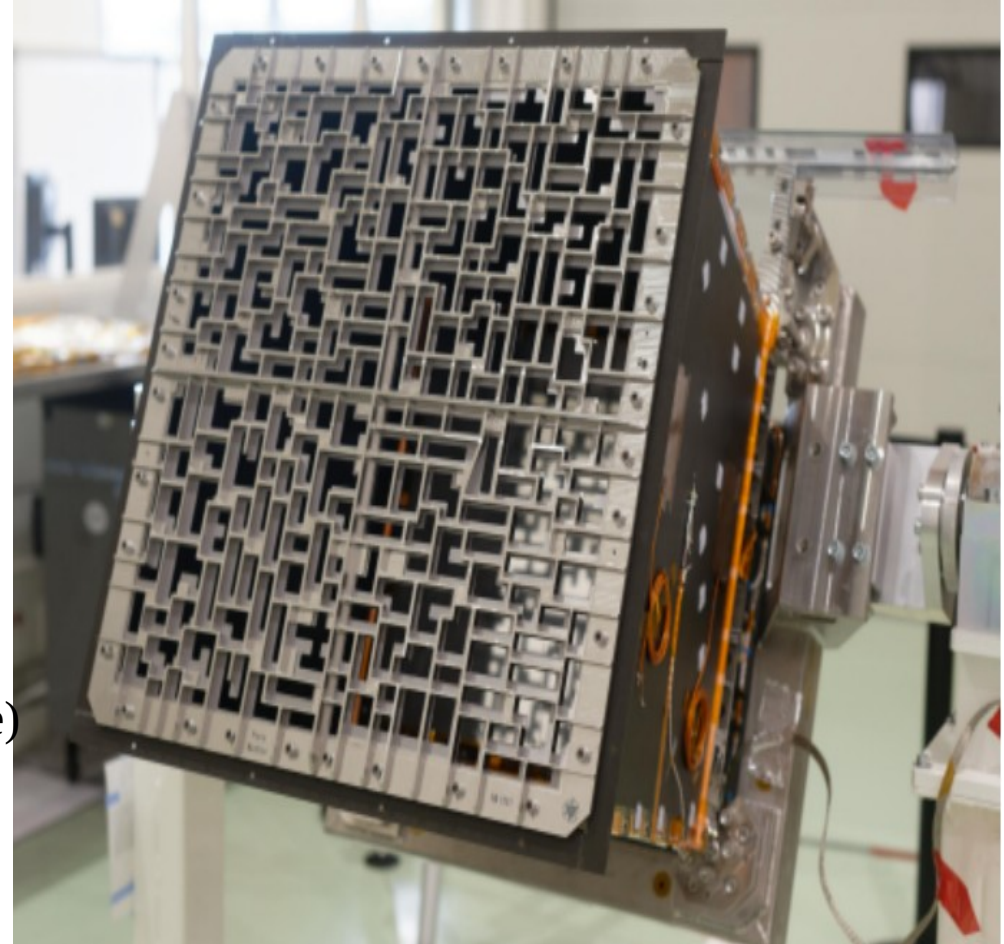


SVOM: ECLAIRs instrument

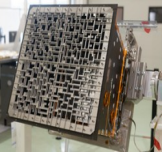


Detector plane

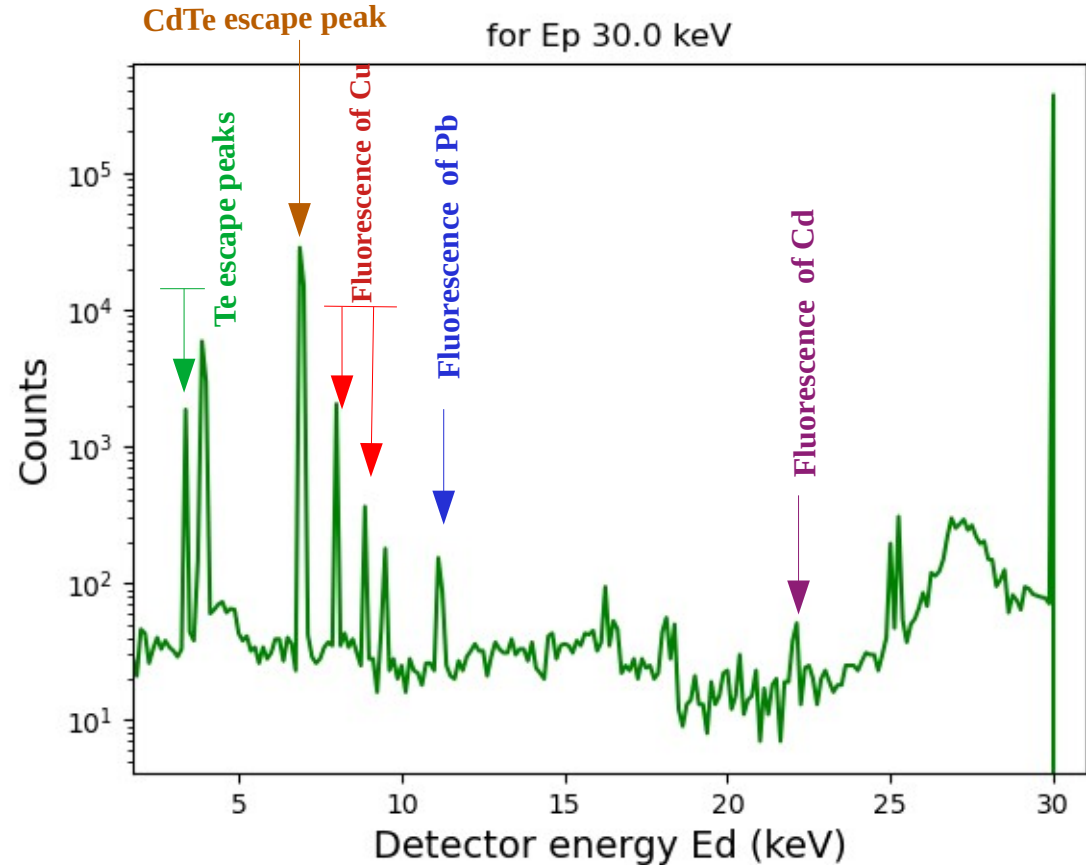
- Consists of 6400 Schottky CdTe (Cadmium Telluride) detectors → low energy threshold of 4 keV
- Pixel size $4 \times 4 \times 1 \text{ mm}^3$
- 80×80 square, a geometrical area of $\sim 1000 \text{ cm}^2$



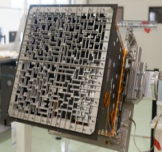
ECLAIRs response matrix function (RMFs)



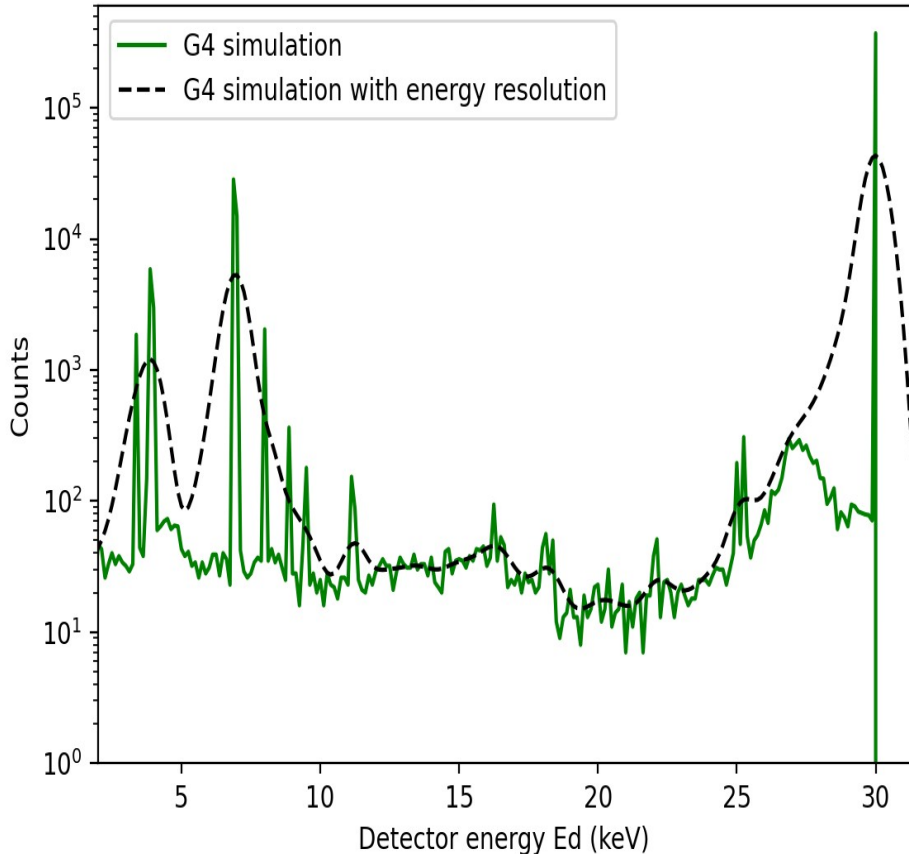
- The numerical simulation takes into account the main physical matter/radiation interactions (e.g. photo-electric effect, X-ray fluorescence, Compton effect)
- Incident photons in the energy range [1.25 keV, 200 keV]
- bin width = 0.125 keV
- Each incident-photon spectrum consists of 10^7 simulated photons
- We consider « single events »
- Source on the on-axis position
- We consider that the Earth is not within the ECLAIRs FoV



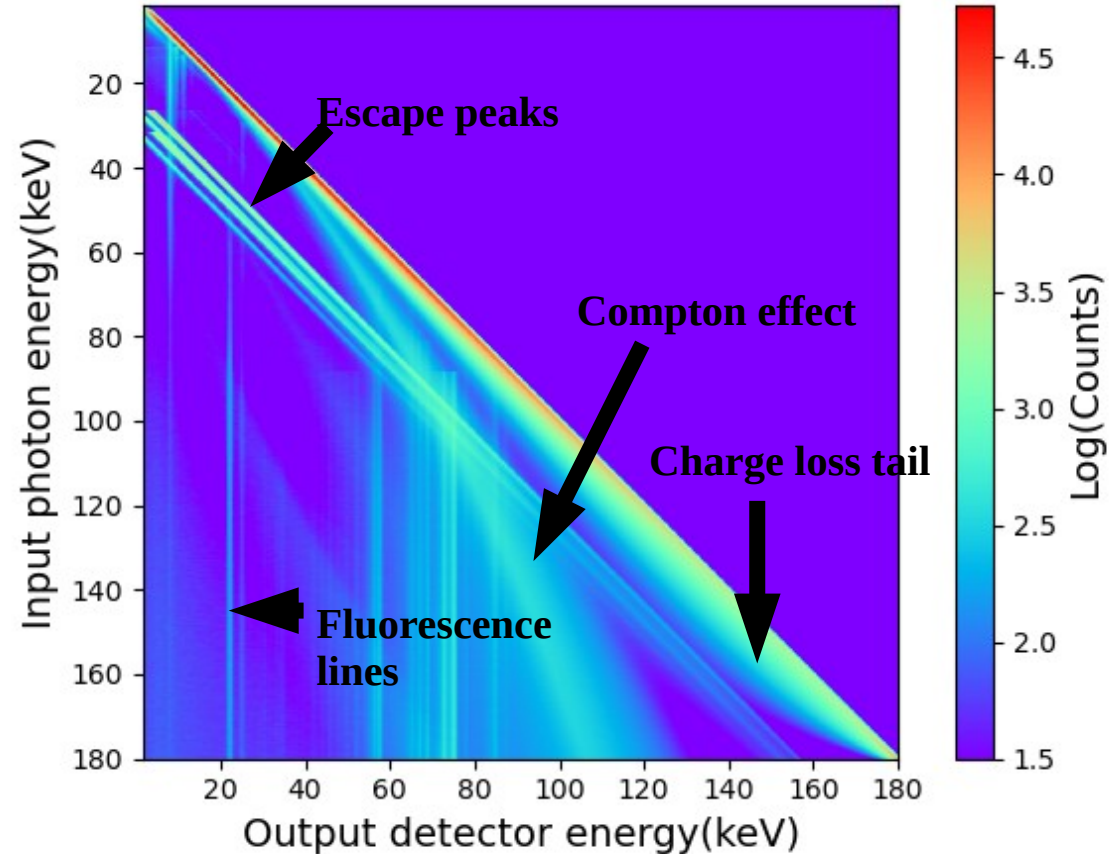
The RMFs using G4 simulation



For Ep 30.0 keV

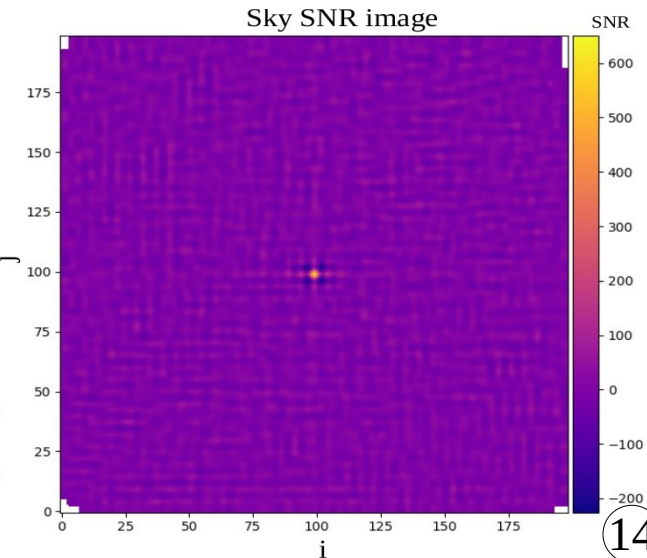
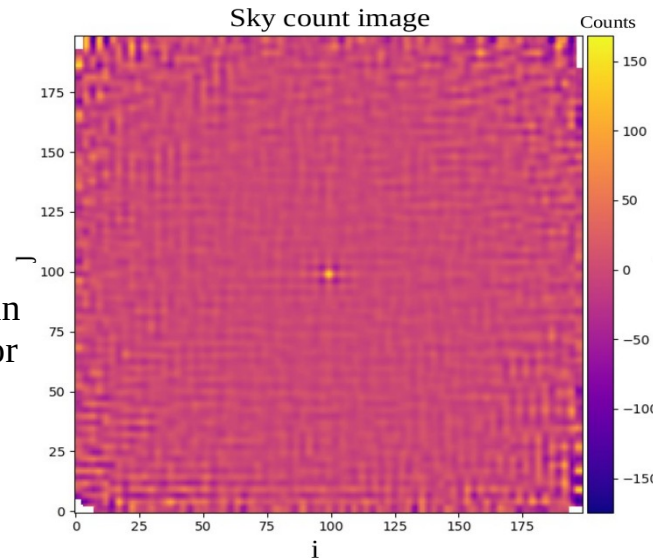
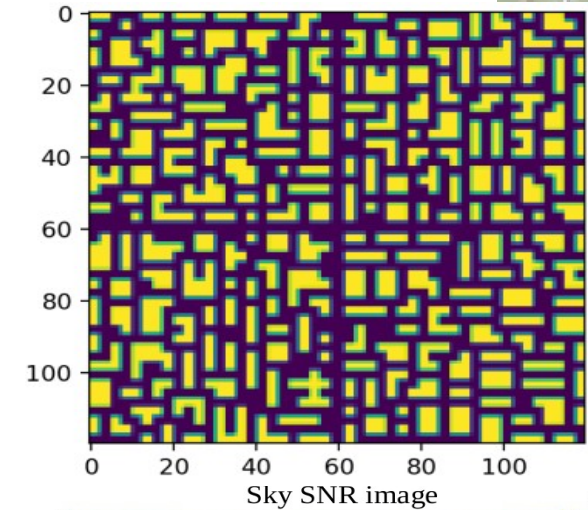


On-axis energy response, simulation with energy dispersion

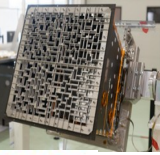


Deconvolution technique

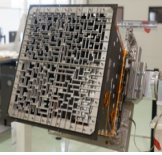
- The source radiation is spatially modulated by a mask of opaque and transparent elements before being recorded by a position sensitive detector.
- The sky-image reconstruction (deconvolution) is based on a correlation procedure between the recorded image and a decoding array derived from the mask pattern.



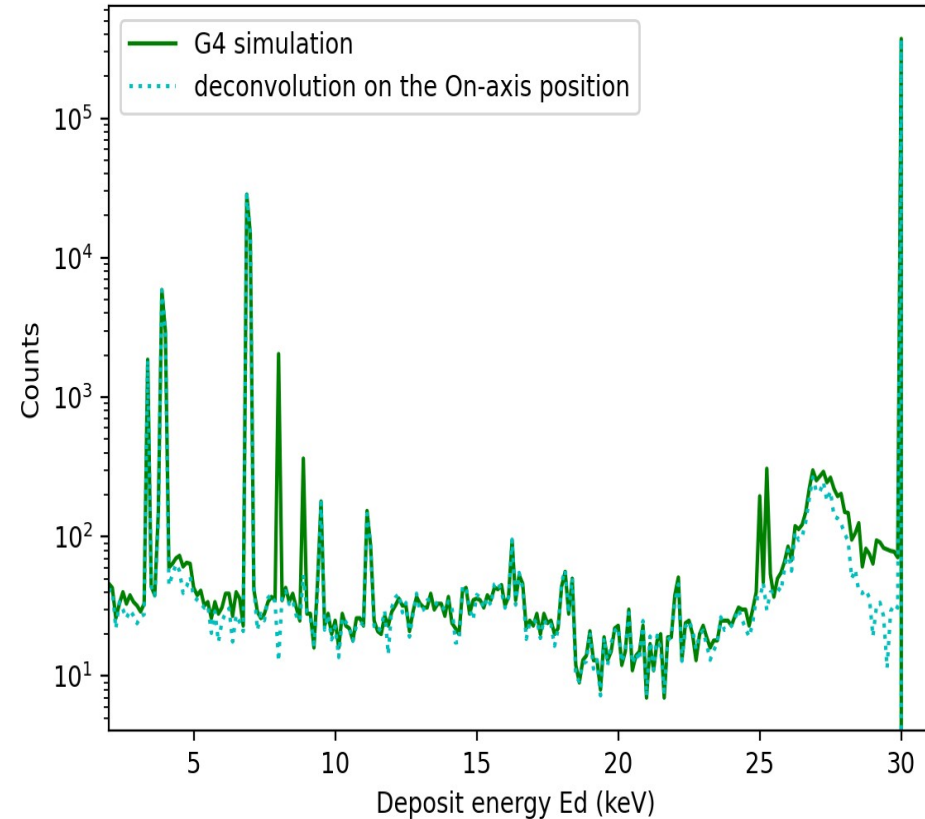
After deconvolution of a shadowgram for an incident-photon of Energy E_p and a detector energy $E_d = 66$ keV



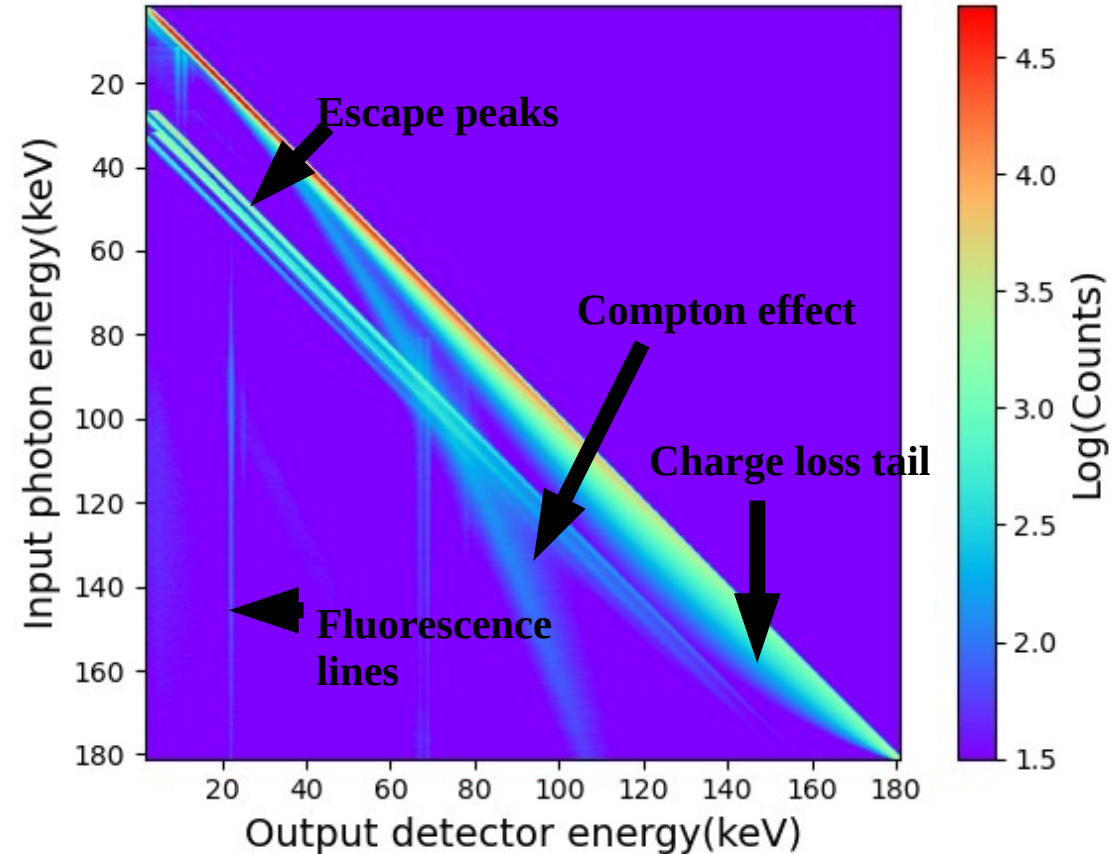
The RMFs after deconvolution

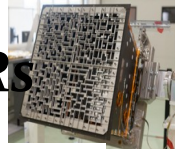


For $E_p = 30$ keV



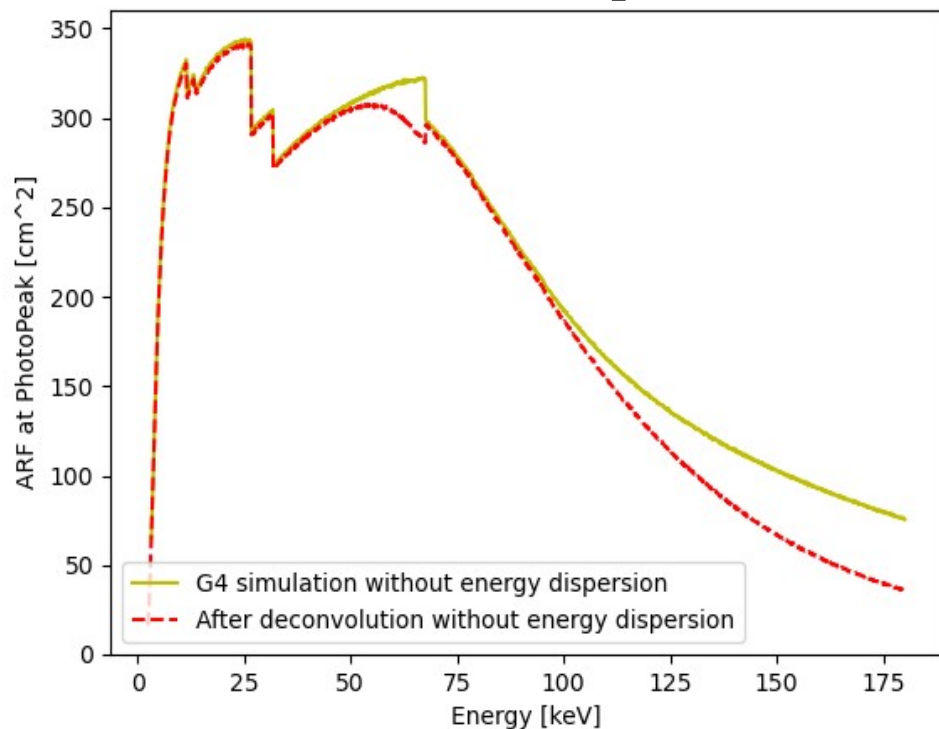
On-axis energy response, deconvolution w energy dispersion



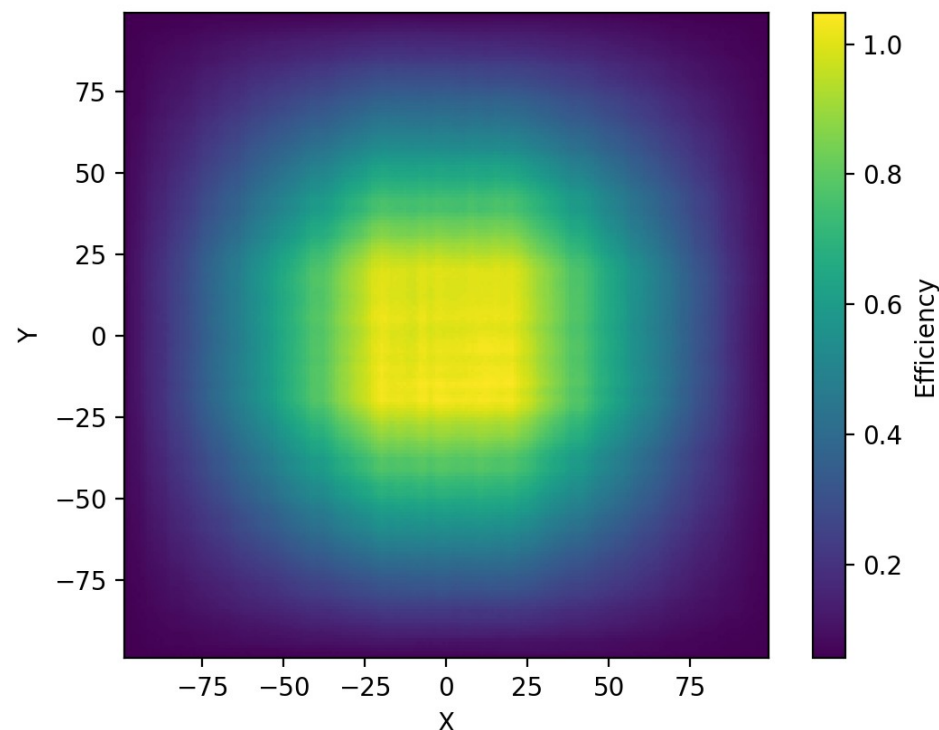


The effective area (ARF) and the Instrument response file (IRF) of ECLAIRs

ARF at Photopeak



IRFs 100 keV

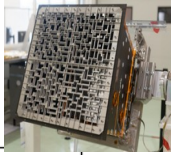


$$IRF(E, \theta, \phi) = ARF(E, \theta, \phi) / ARF(E, 0, 0)$$

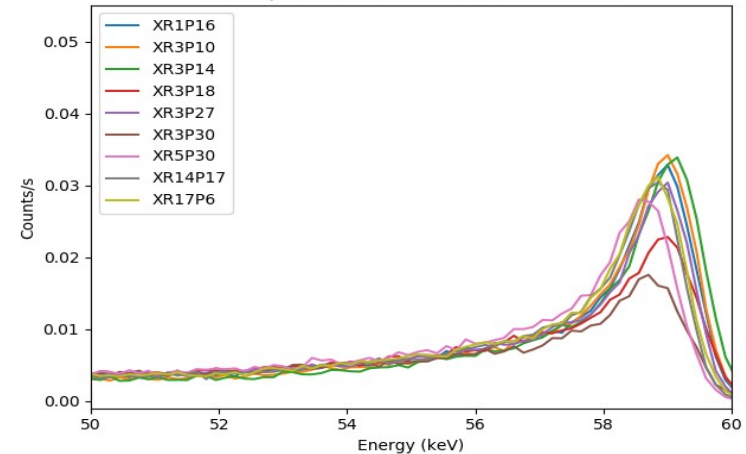
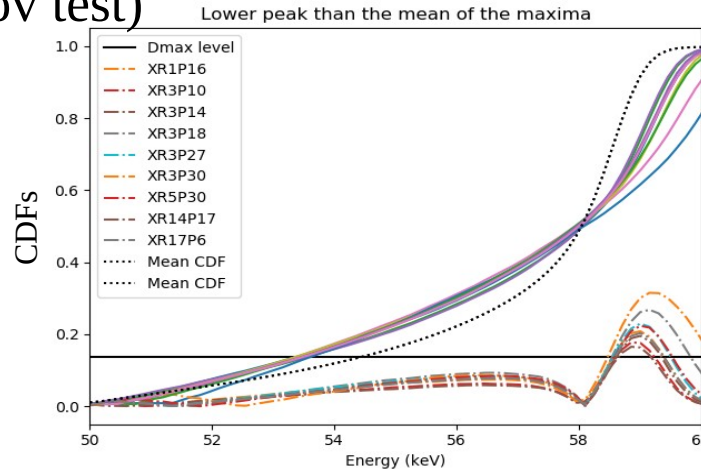
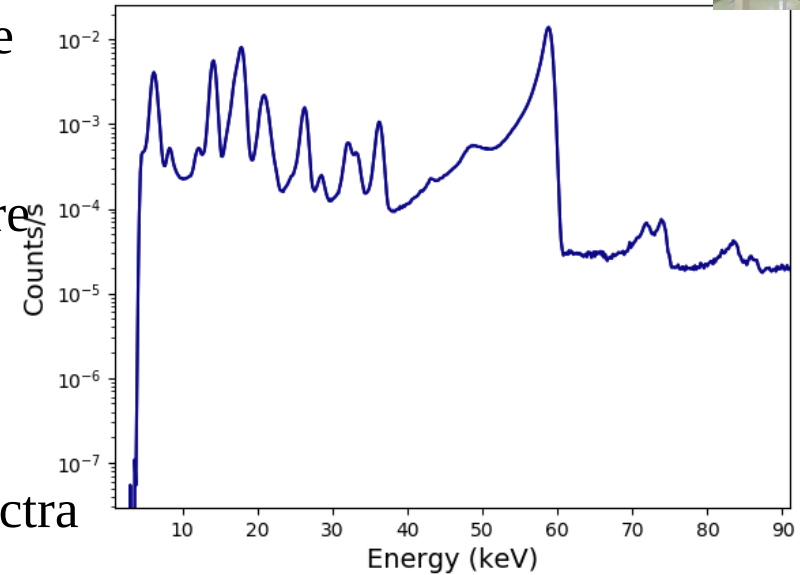
- Consistency between the simulated and reconstructed ARFs at low energy
- The mask becomes transparent at high-energies. The events that passed the opaque pixels are considered as Bkg events and excluded by the deconvolution technique

Performance of the *SVOM ECLAIRs* instrument

ELS1 -31/-32°C



- Participate to the calibration tests of the detection plane and on the whole camera.
- Monitor the spectral response of ECLAIRs and measure all the parameters that can affect the final response of the telescope (temperature, angle, ...)
- Identify groups of pixels with common characteristics. Check the characteristics of pixels with anomalous spectra (Kolmogorov Smirnov test)

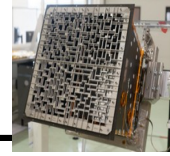


Preparation of early science with CTA

- *Working on pipeline development (ctapipe/protopipe)*
 - Worked within LSST DESC collaboration
- *Validation/Improvement of the LST1-MAGIC pipeline*
 - Contributed to inter-calibration between GBM-LAT Fermi instruments (ongoing work with SVOM)
- *Estimate the sensitivity of the IRFs with respect to the observation angle*
 - Built the IRFs using G4 simulated data
- *Data simulation, statistical analysis, modeling*
 - My work with Fermi, CTA, SVOM
- *Use deep-Learning technics*
 - Training CFHT dataset for photo-z measurement using tensorflow (2020)
 - Followed a part of the Fidle training about: introduction to deep learning

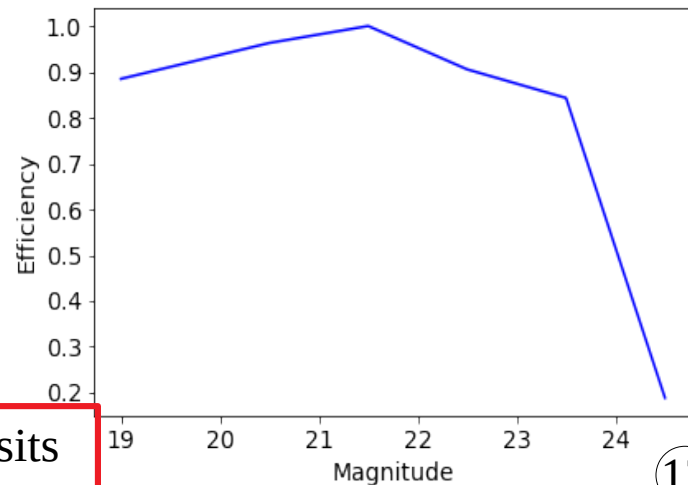
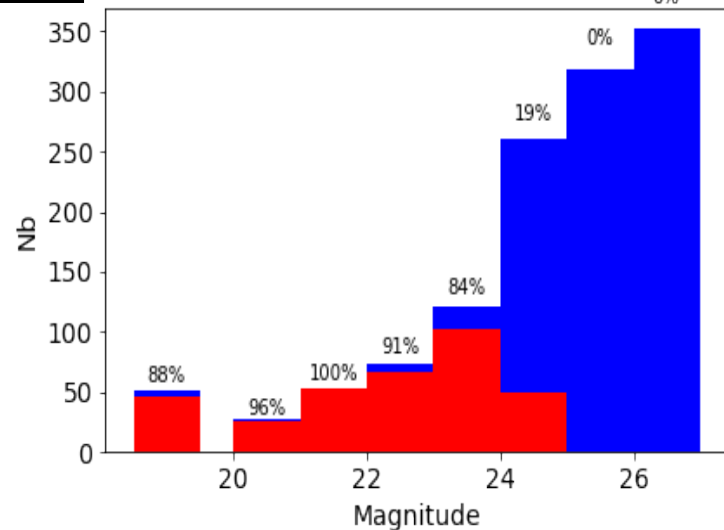
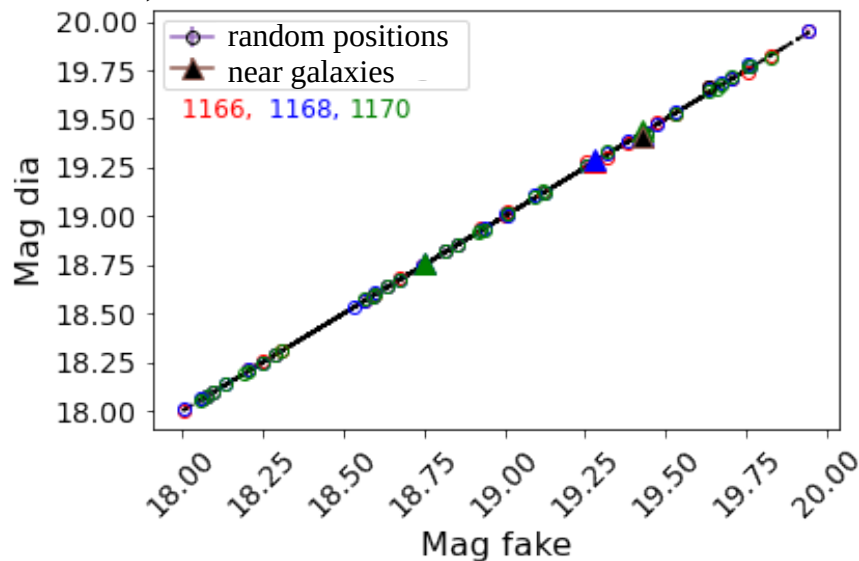
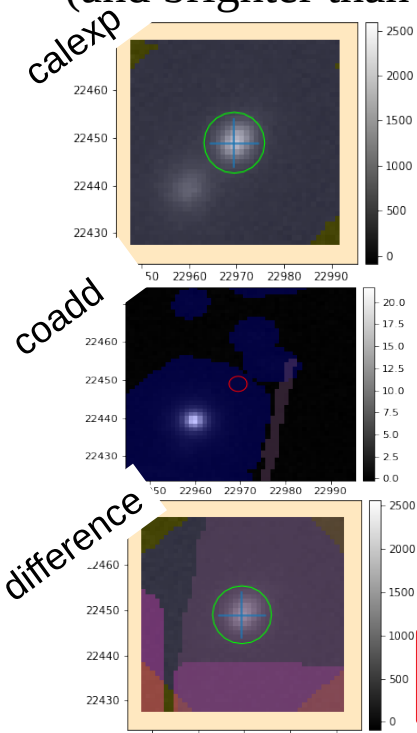
Thank you

Back-up slides



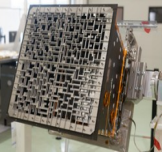
Detection efficiency

- Input 52 fakes with $18 < \text{mag} < 20$
- 46 among 52 are retrieved with angular distance < 0.12 as
- Only 3 fakes are associated with galaxies
(and brighter than their hosts)

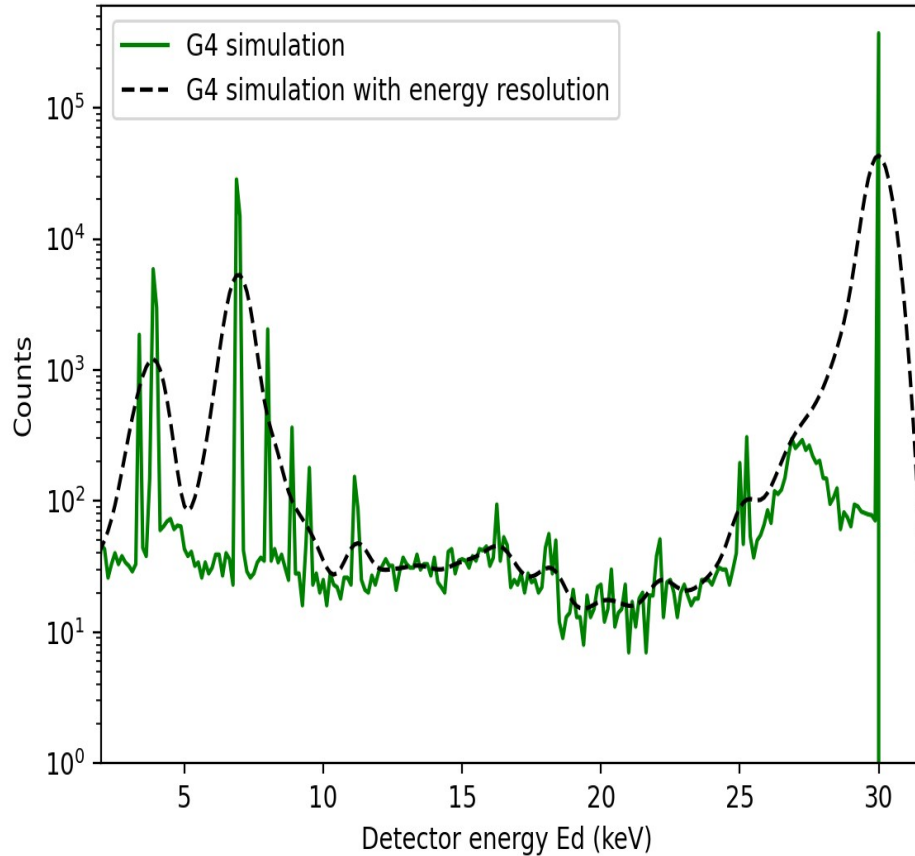


Well retrieved magnitudes for all fakes in the 3 visits

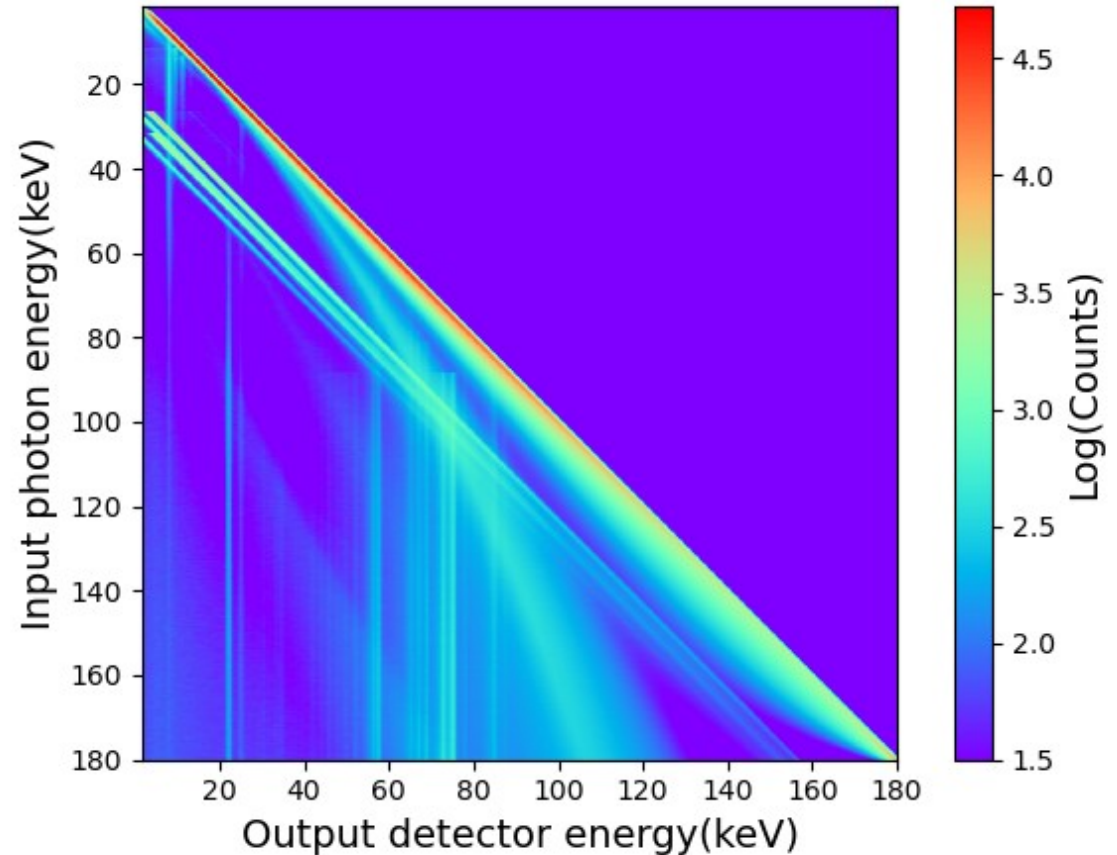
The RMFs using G4 simulation



For Ep 30.0 keV



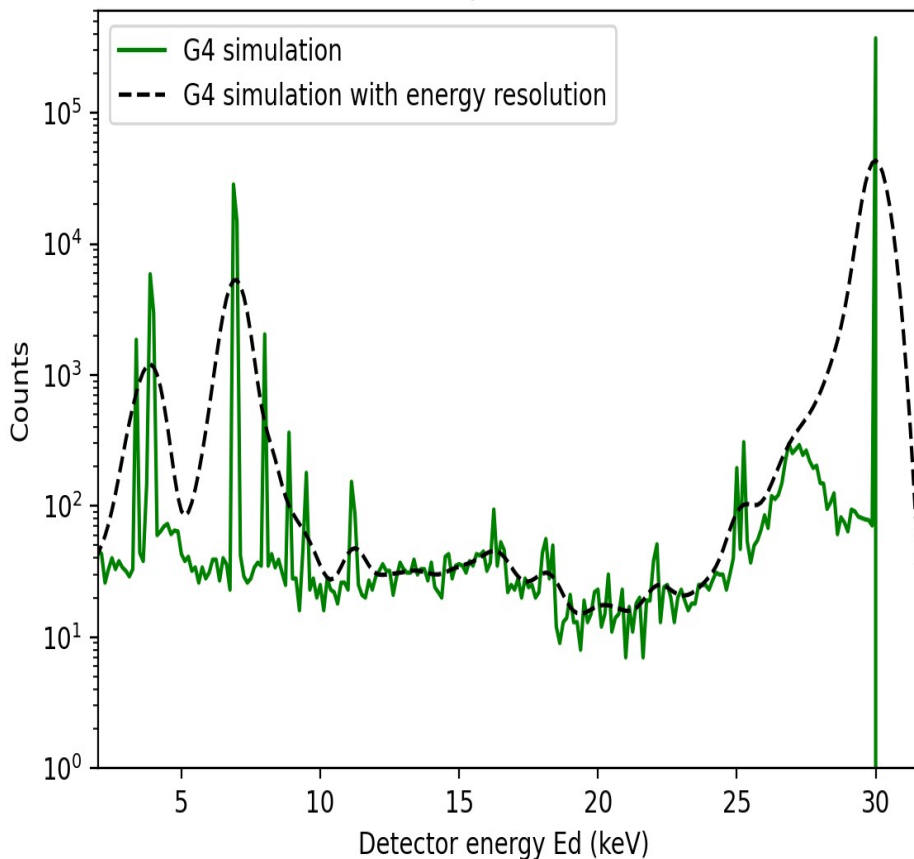
On-axis energy response, simulation with energy dispersion



The RMFs using G4 simulation



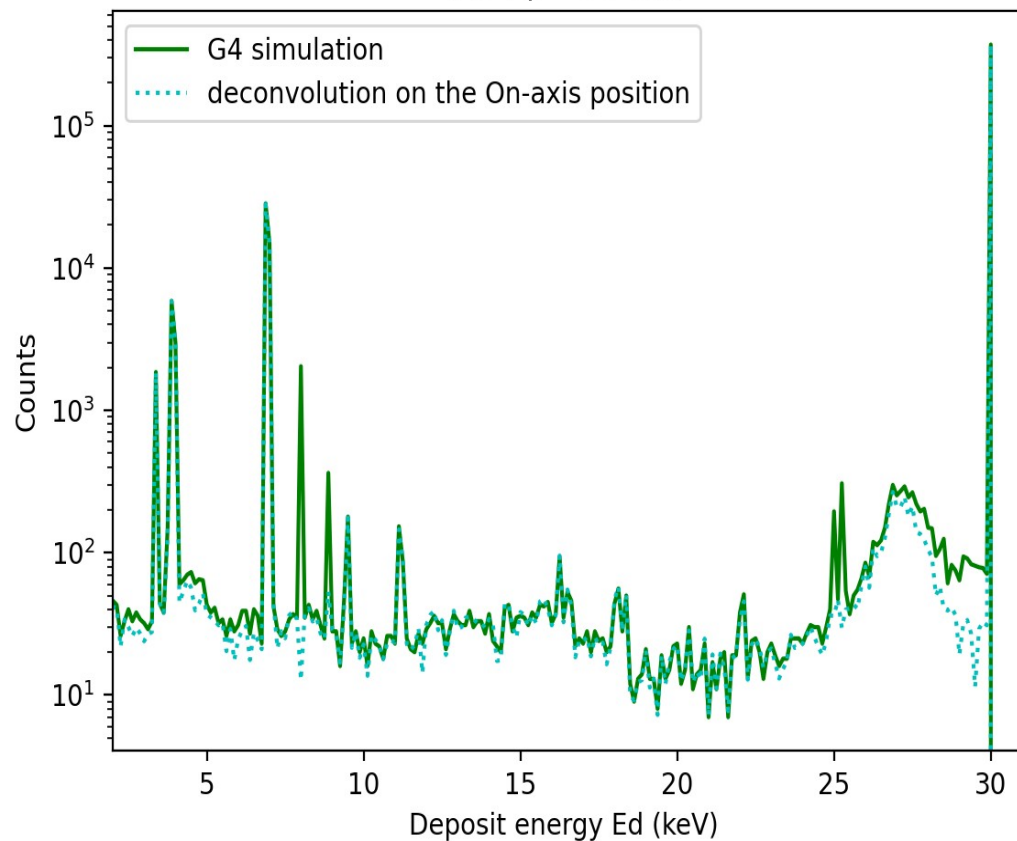
- We use the Helmer-Lee phenomenological function
- Validated using the calibration data of the prototype with the sources (^{241}Am , ^{55}Fe , ^{133}Ba , ^{57}Co)
For Ep 30.0 keV



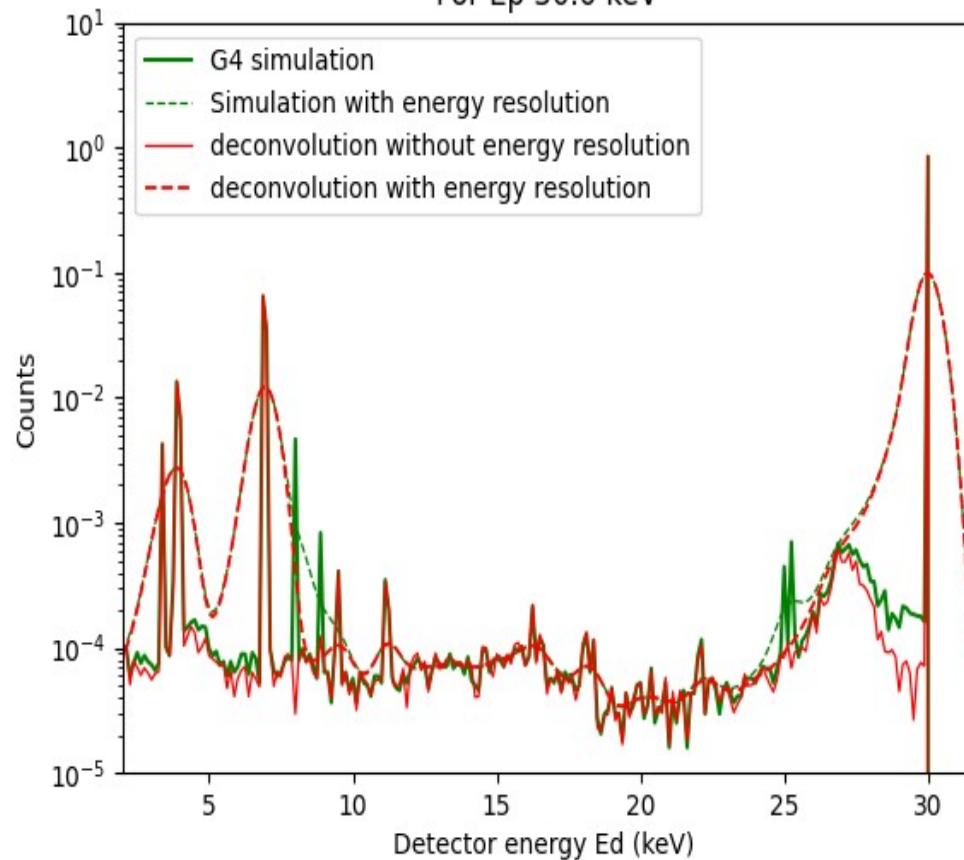
The RMFs after deconvolution



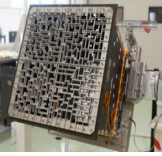
For $E_p = 30$ keV



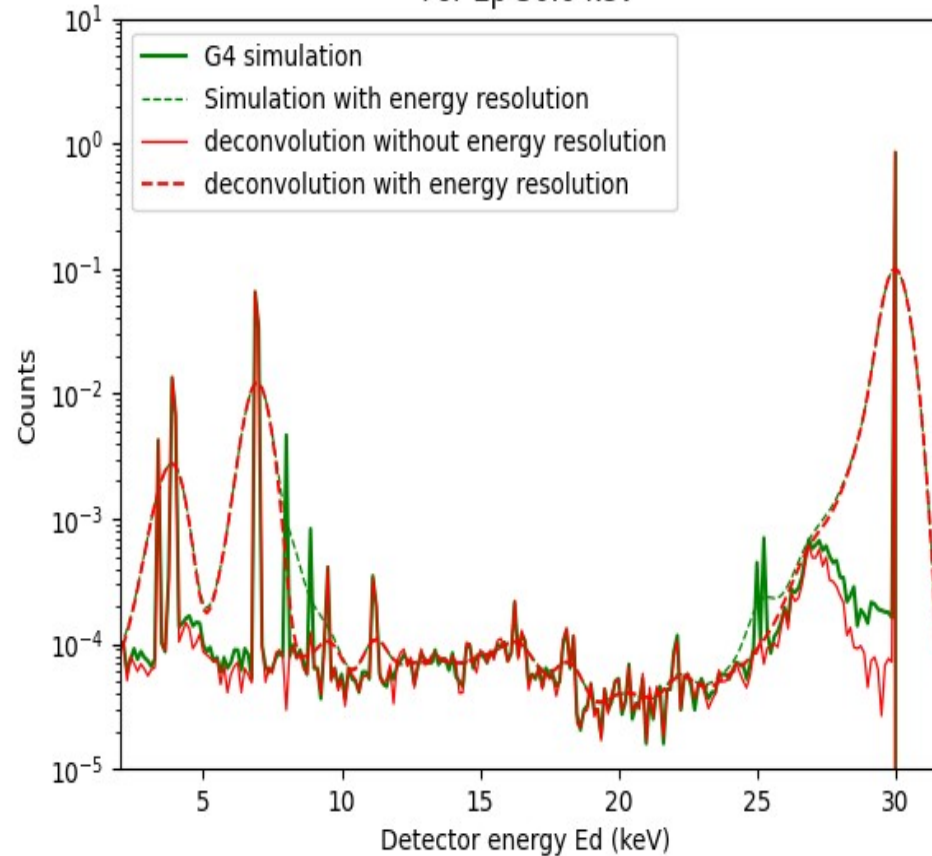
For $E_p = 30.0$ keV



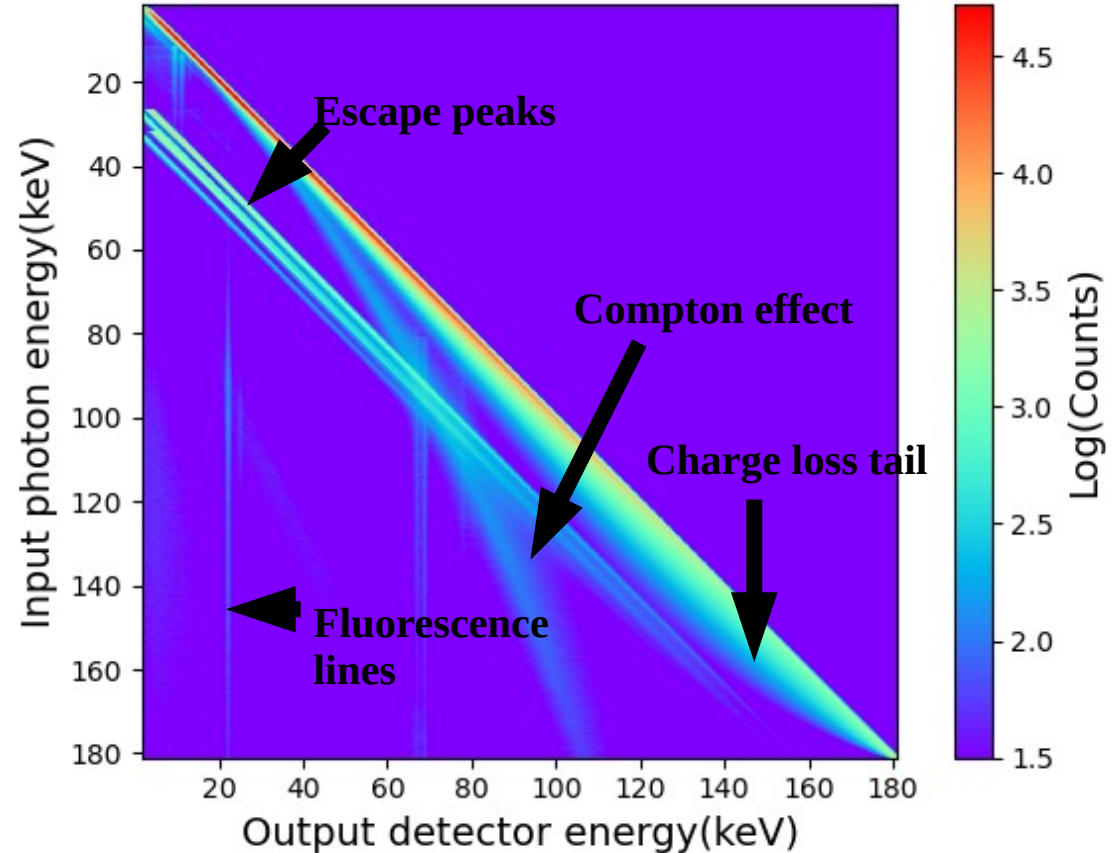
The RMFs after deconvolution



For Ep 30.0 keV



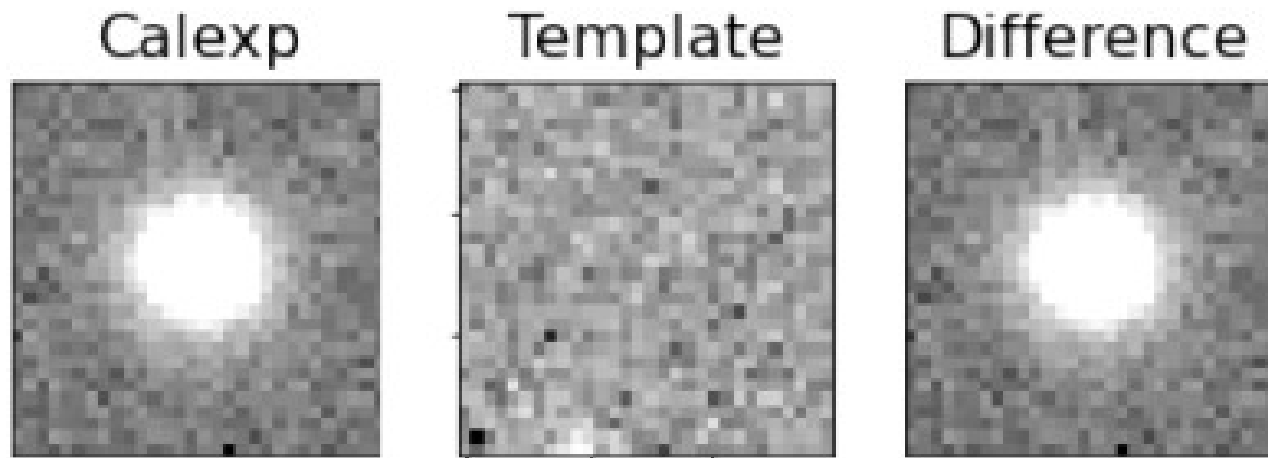
On-axis energy response, deconvolution w energy dispersion



Validation of the DIA pipeline

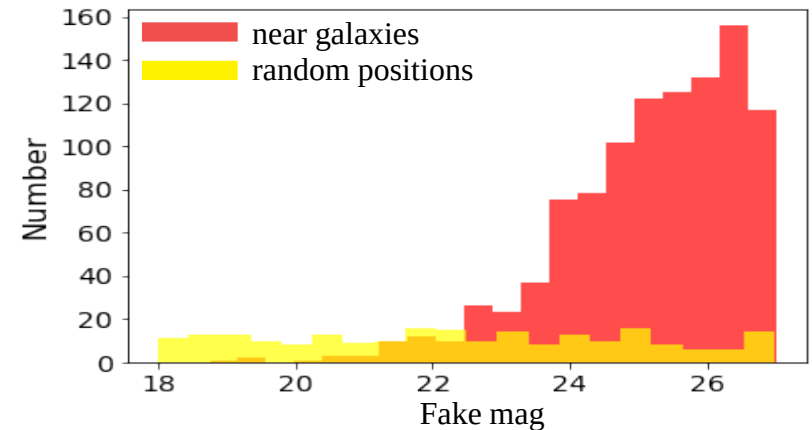
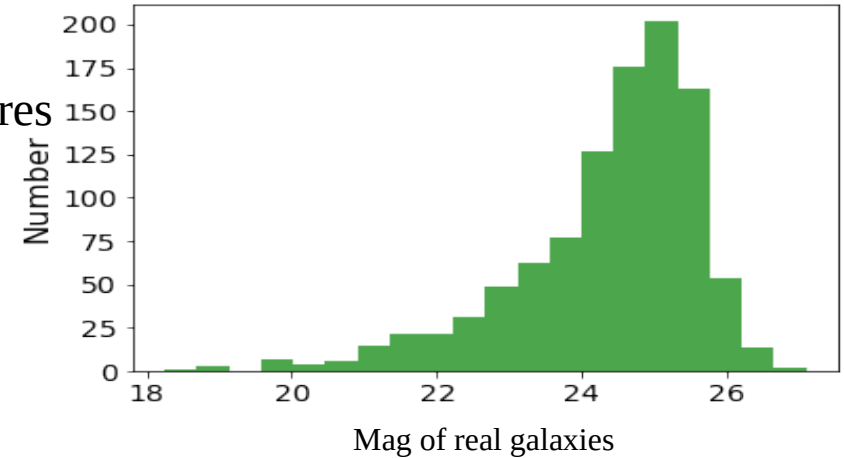
Goal: evaluate the performance of the difference imaging pipeline

- Use the fake injection technique:
 - Inject fakes onto science images (in calexps)
 - Subtract images from a templates of stacked exposures (coadd)
 - Check the efficiency of detection

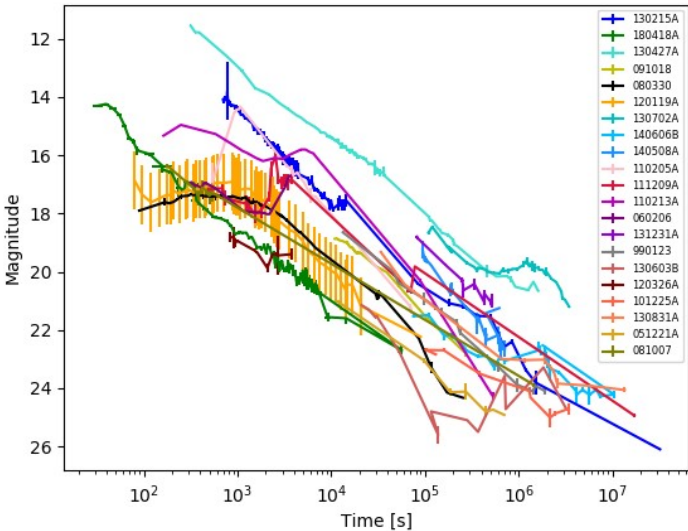


Fake injection

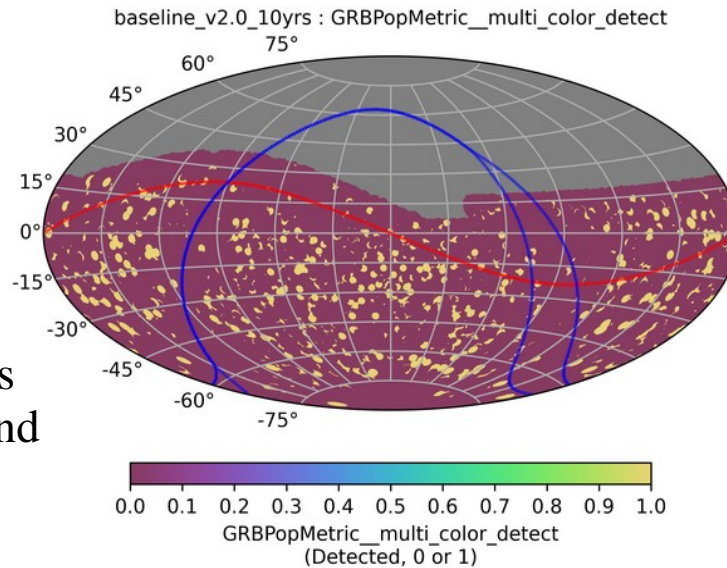
- Inject fakes onto science images (in calexps)
- Subtract images from a coadd templates of stacked exposures
- Check the efficiency of detection
- Generate fakes associated to a galaxy (on 10% of randomly chosen galaxies)
- Gaussian distribution: choose a random position according to the galaxy radius
- Flat distribution to choose mag ([galaxy mag -1, 27])
- Generate 2% fakes with random positions (without galaxy association)
- Check the detection within the magnitude intervals



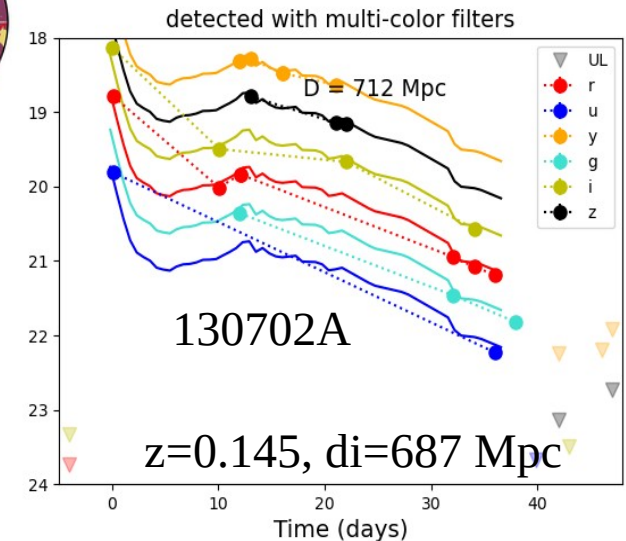
GRB detection with LSST



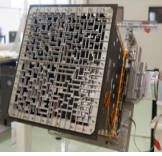
- LCs uniformly distributed in volume
- Use the baseline cadence
- Define a metric (Metrics Analysis Framework)
- Define the detectability criteria



➔ Construct the output LCs of LSST observations and compare them with the real LCs

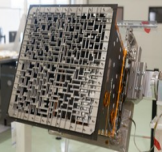


Identify bad pixels: Kolmogorov test

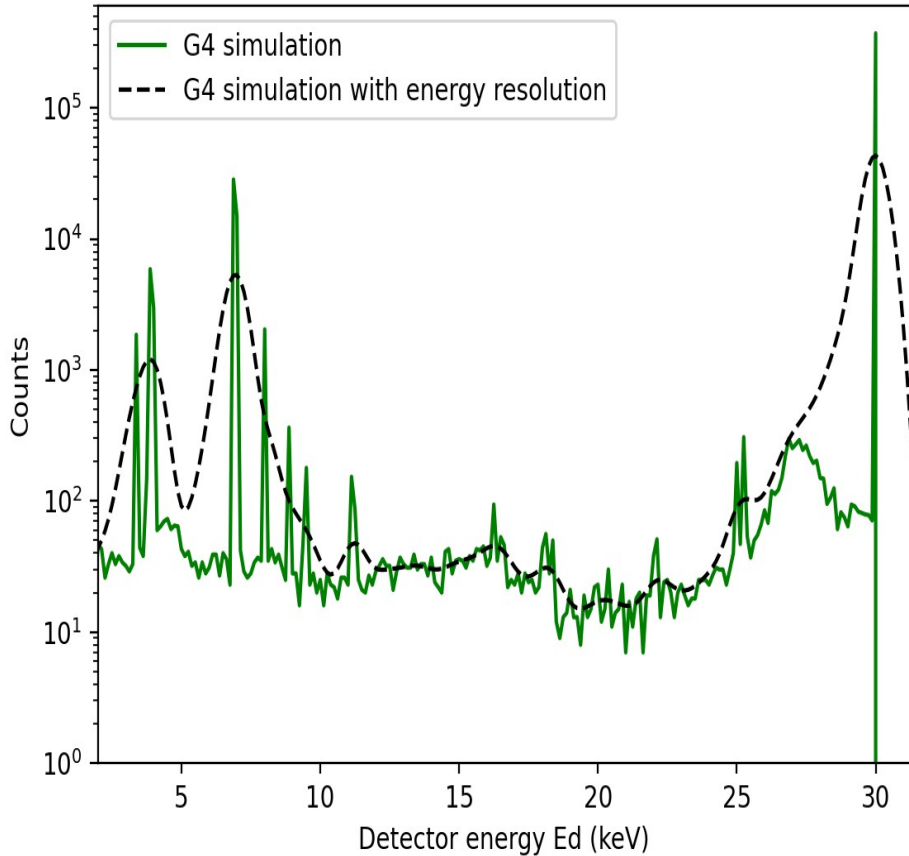


- Calibrate the spectra and find the mean spectrum per sector
- For each sector, shift all CDFs and the CDF of the mean spectrum at the peak energy of the source spectrum
- Dmax: the maximal distance between the CDFs of pixels and the CDF of the mean spectrum within the same energy bins
- Dmax is compared to Dmax level at 95 % of confidence level ($\alpha=0.05$)
- Compare the max counts/s for excluded pixels wrt to the mean counts/s for all pixels

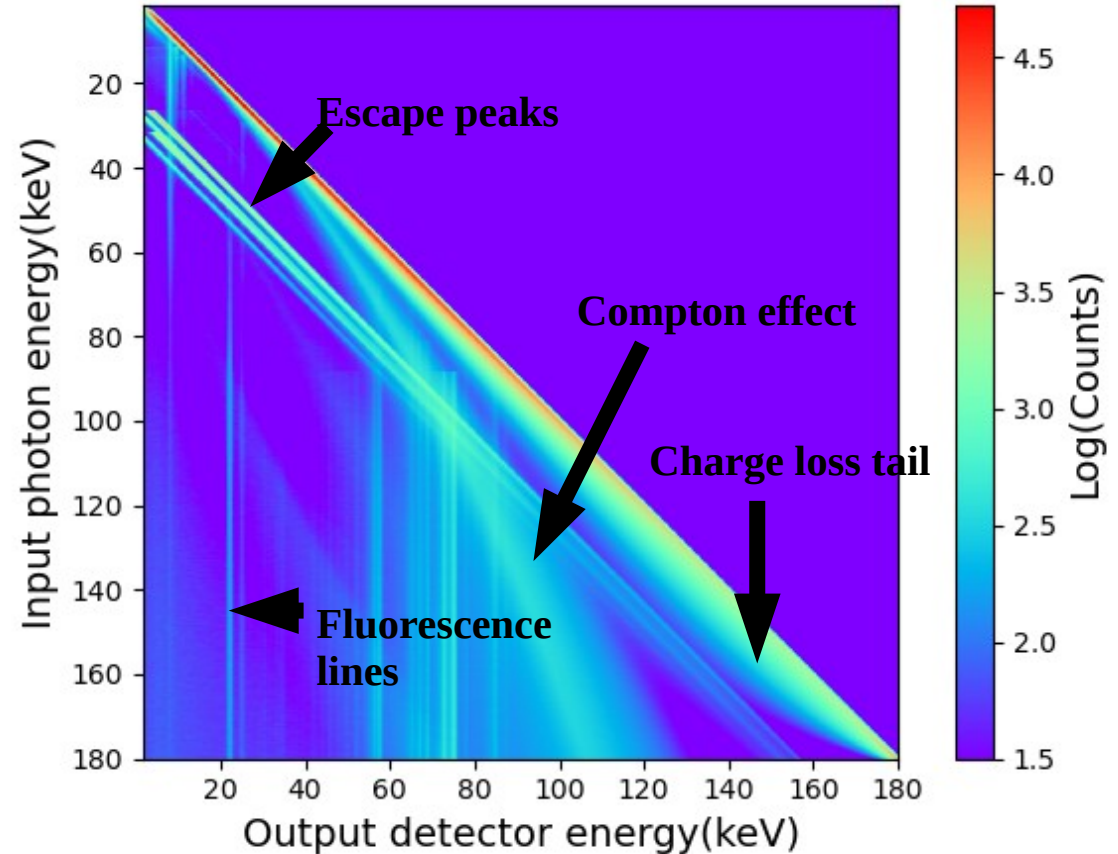
The RMFs using G4 simulation



For Ep 30.0 keV



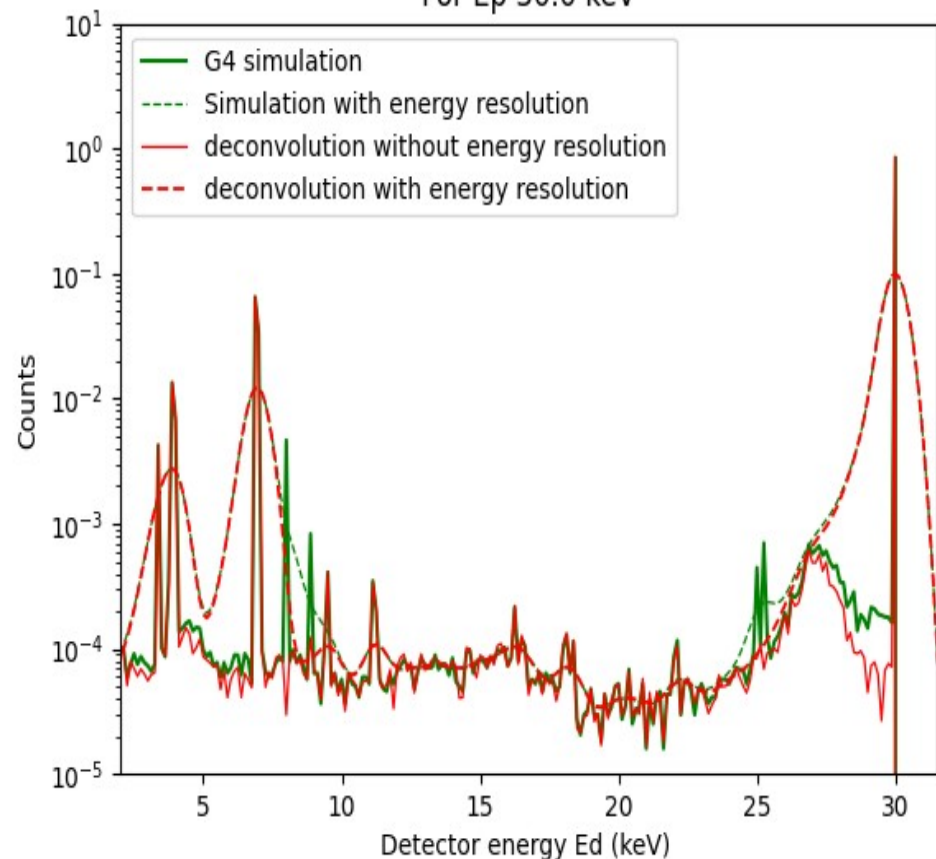
On-axis energy response, simulation with energy dispersion



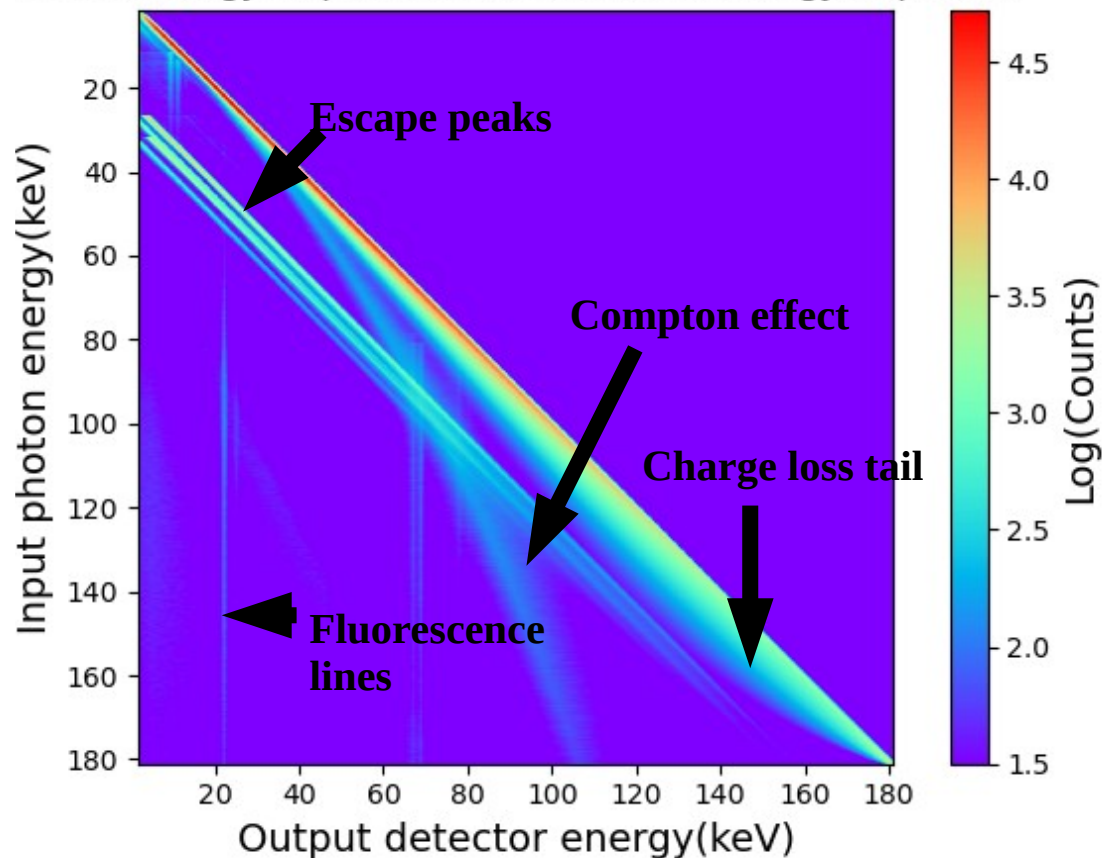
The RMFs after deconvolution



For Ep 30.0 keV



On-axis energy response, deconvolution w energy dispersion



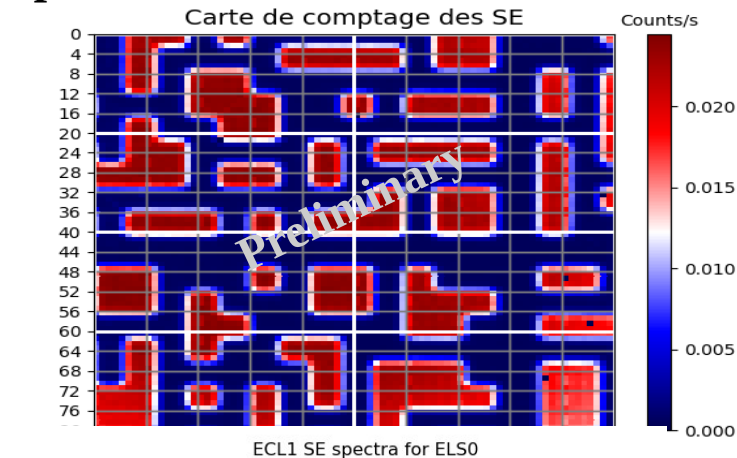
Performance of the instrument



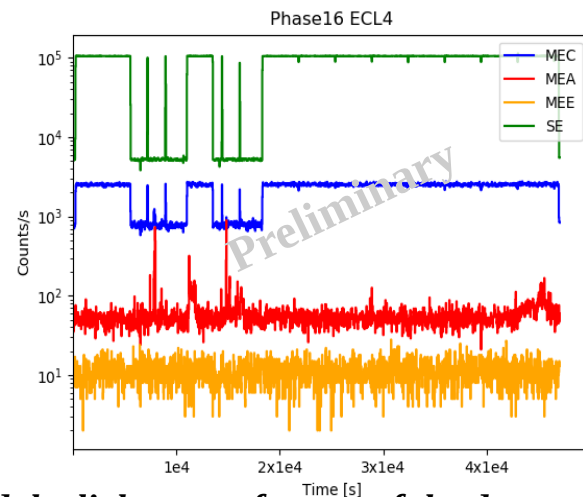
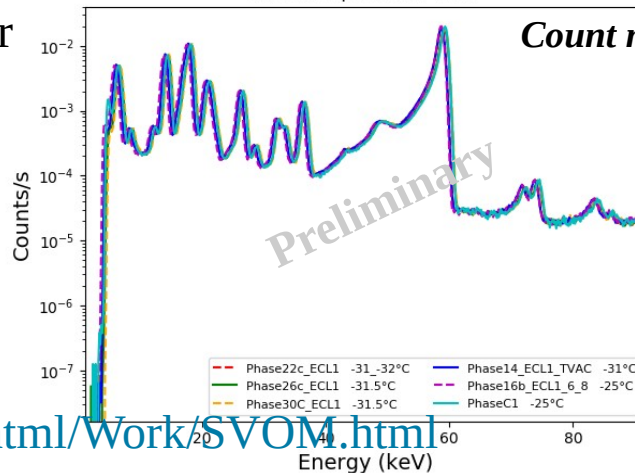
Several configurations to check the performance of the instrument :

- Different radioactive sources
 - Gamma-rays (Am, Fe, ..)
 - X-rays (Sc, Cr, ..)
- Source at different angles
- A set of different temperatures
- SAA or nominal modes

- Check the detection efficiency per sector and for each pixel
- Construct the spectrum using the gain-offset of each of the pixels (Channel → energy relation)

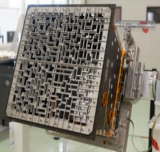


ECL1 SE spectra for ELS0



Count map and the light curve for one of the datasets

Energy dispersion function



- We use the Helmer-Lee phenomenological function

S14 function, Helmer et Lee (, RD-2)

It consists of the three components : f1 : Gaussian
f2 : exponential and f3 the error function
It has 6 parameters to fit

$$S_{14}(E, \mu, \sigma, p_4, p_5, p_6, p_7) = \frac{f_1 + f_2 \times f_3}{1 + 2 p_4 \times p_5 \times \exp\left(\frac{\left(\frac{\sigma \times p_6}{p_5}\right)^2}{2}\right)}$$

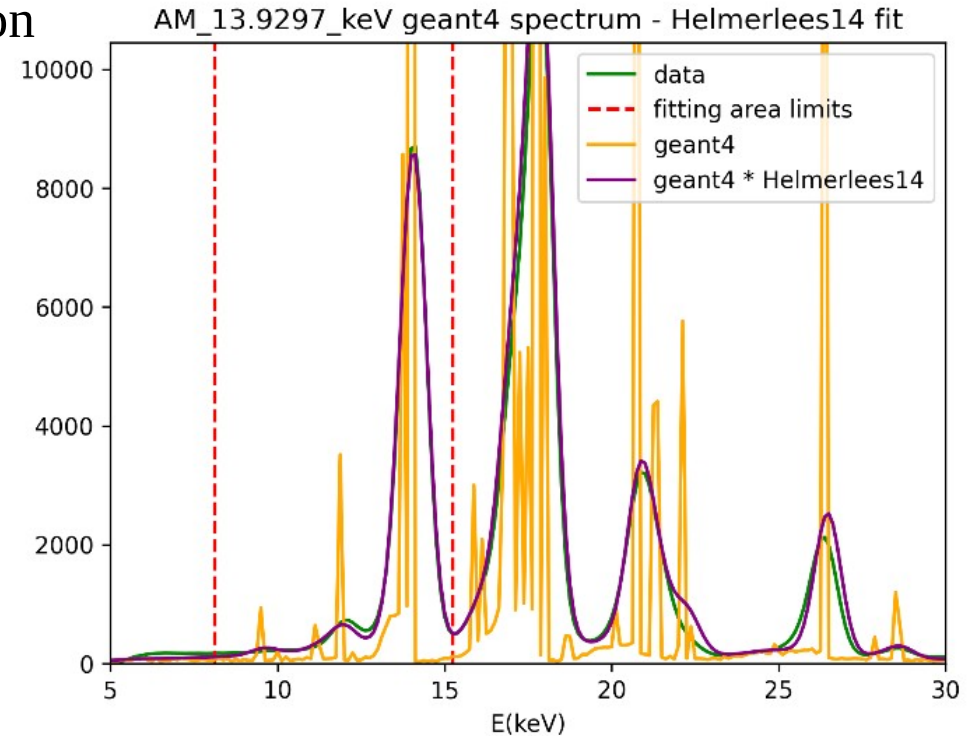
where

$$f_1 = G(x, \mu, \sigma) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$f_2 = p_4 \exp\left(\frac{E - \mu'}{p_5}\right) \text{ et } \mu' = \mu + p_7$$

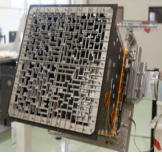
$$f_3 = \text{erfc}(u) \text{ et } u = \frac{x - \mu'}{v} + \frac{v}{2 \times p_5} \text{ avec } v = \sqrt{2} \sigma \times p_6$$

$\sigma, \mu, p_4, p_5, p_6, p_7$ are the fit parameters



- Was validated using the calibration data of the EQM prototype with the sources (^{241}Am , ^{55}Fe , ^{133}Ba , ^{57}Co)

Deconvolution technique

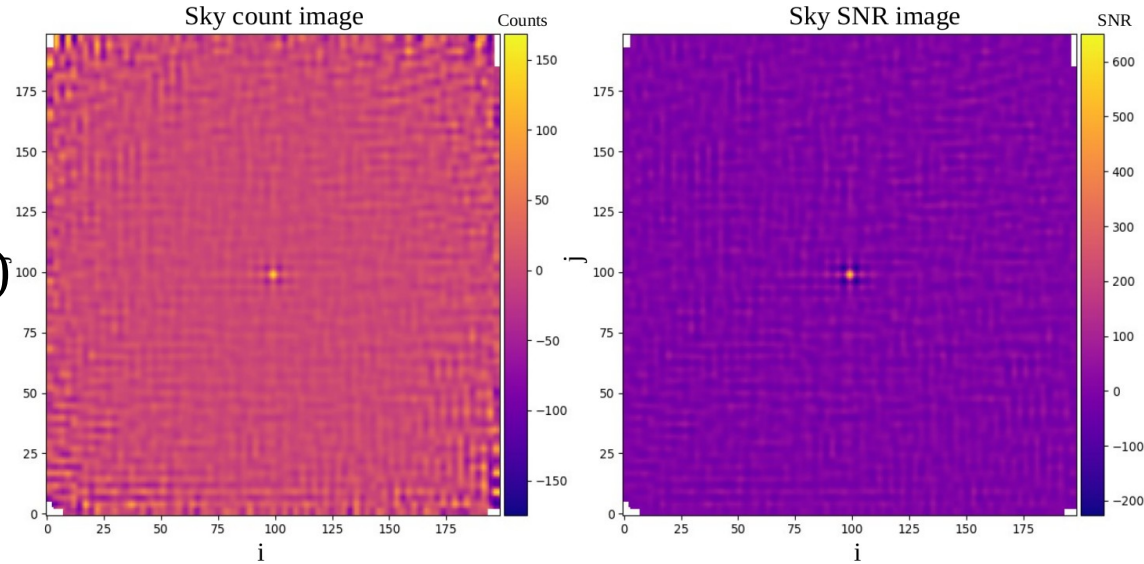


- Build shadowgrams (i, j, E_d, E_p) for i, j the pixel coords, E_d deposited energy (channel) and E_p the incident energy

- For each E_p deconvolve the images in all bins of deposited energies (E_d)

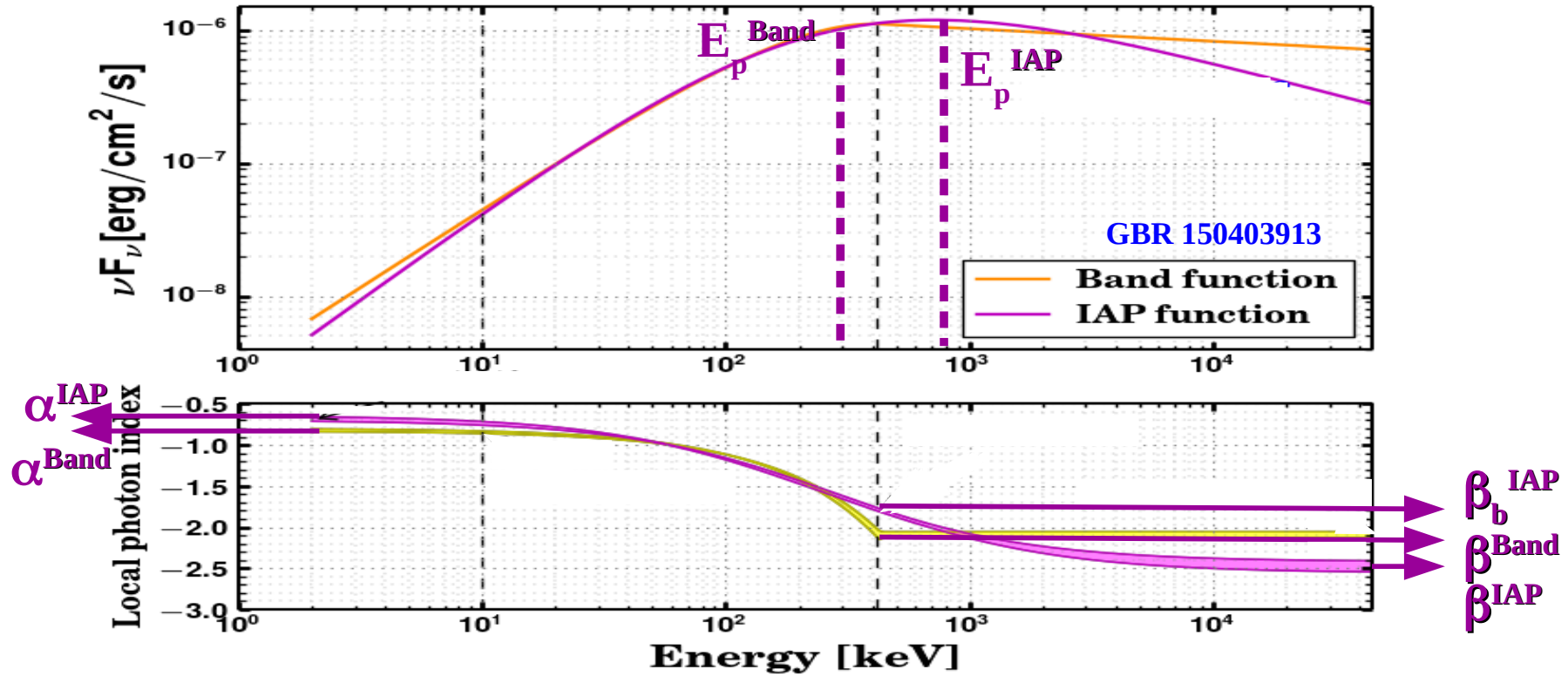
- Effective area of a pixel = 0.147 cm^2
- Mask opening = $\Sigma \text{mask} / \text{mask surface}$ (120^2 for the binned mask)

$$\text{Counts}(E_d) = \frac{\text{counts}_{\text{on-axis}} \times 6400 \times \text{eff area}_{\text{pixel}}}{\text{mask opening}}$$



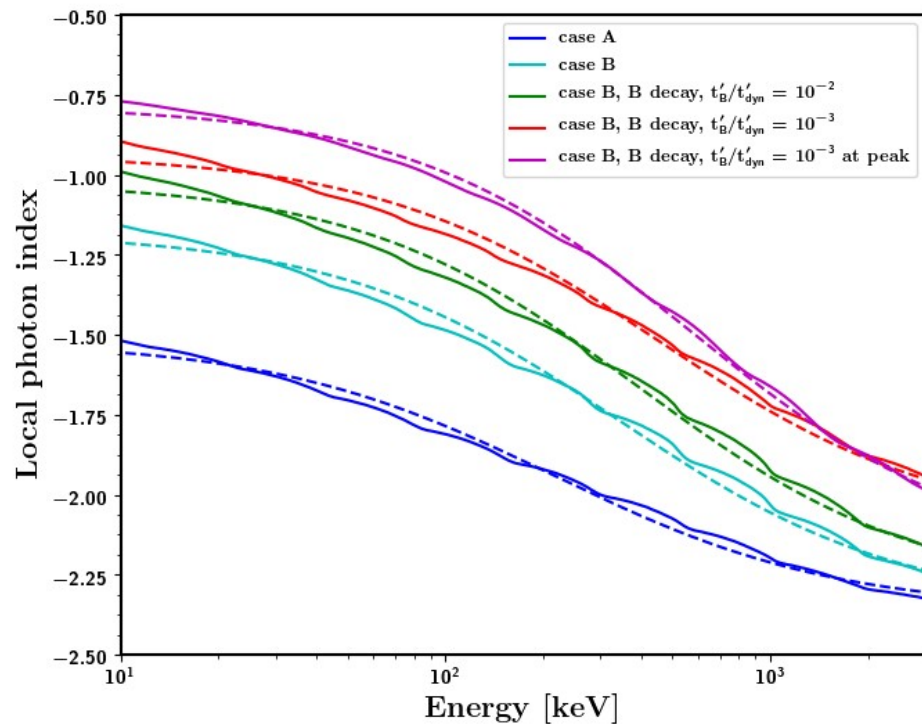
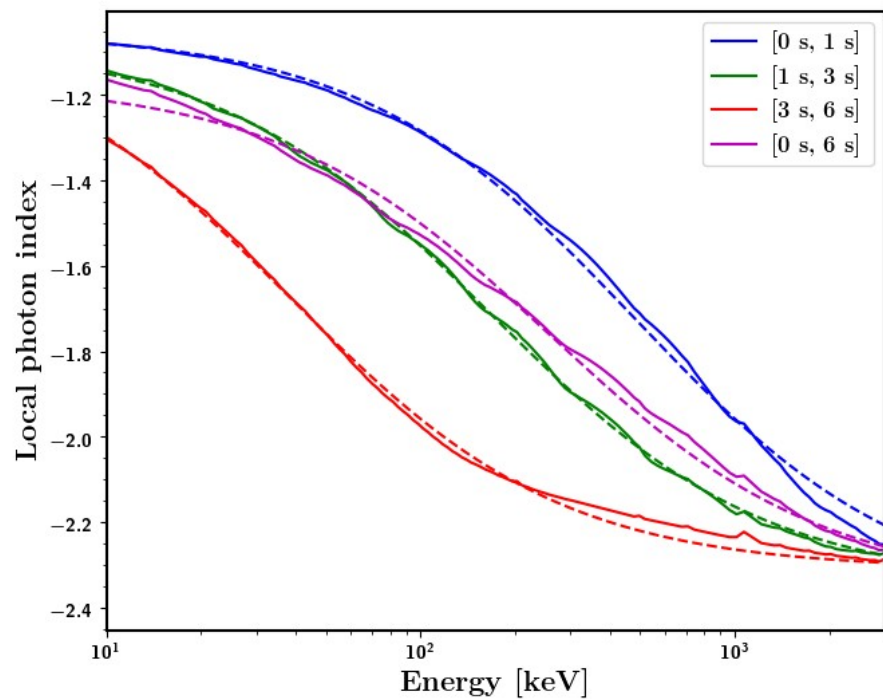
The sky-count image and the sky-Signal to Noise Ratio image after deconvolution for a shadowgram off an incident-photon of energy = 66 keV

Comparison of typical IAP and Band spectra

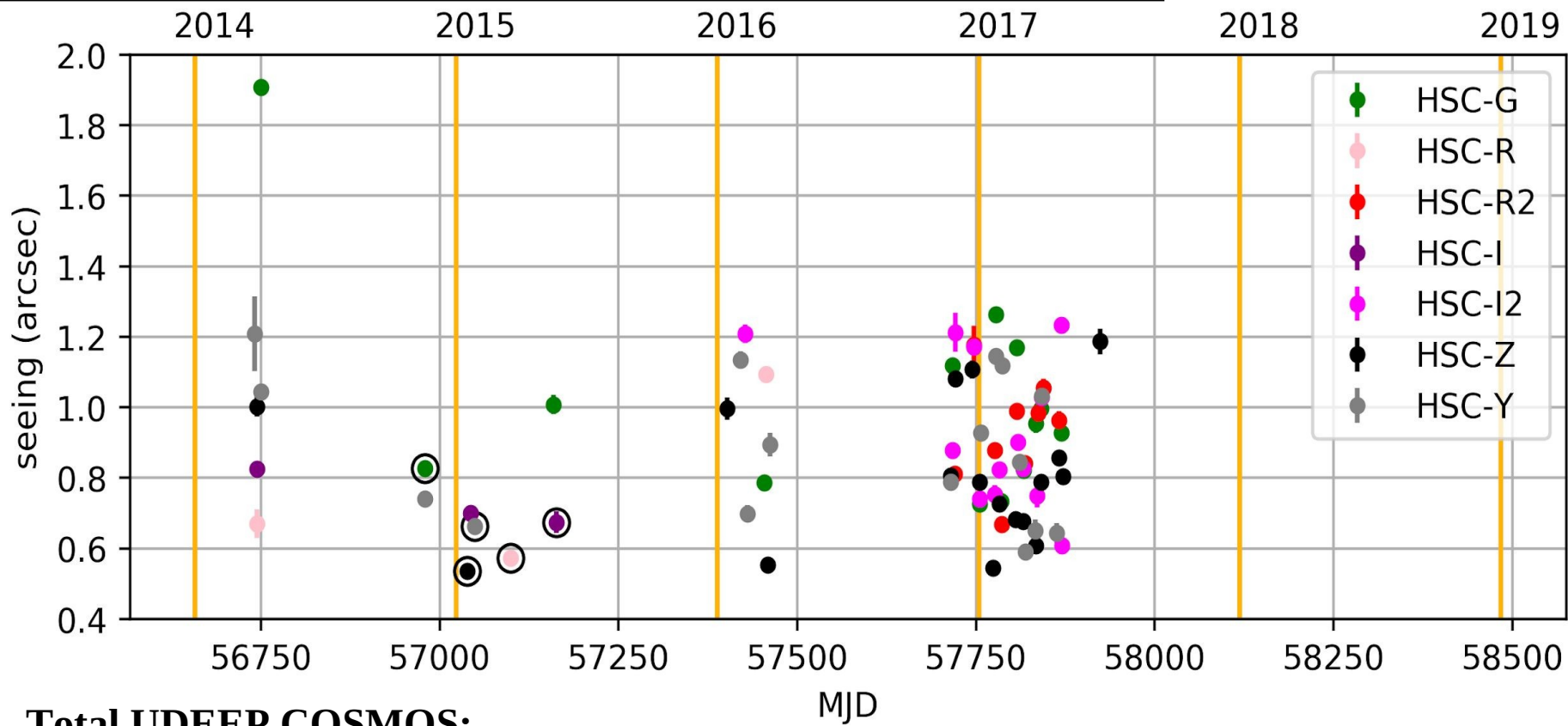


- The **IAP** spectrum is wider around the peak energy than the **Band** spectrum, but narrower when observed over a wider energy range.
- The same trend is observed for the GRB sample as for the synthetic bursts

Comparison of typical IAP and Band spectra



DATA



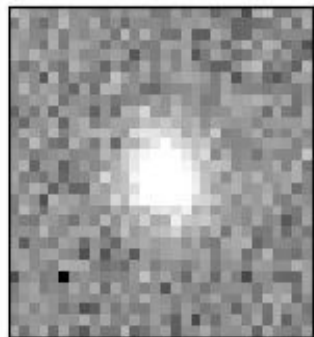
Total UDEEP COSMOS:

54 pointings

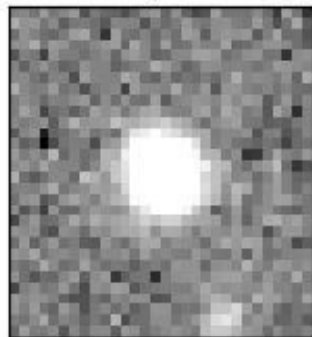
642 visits

Results

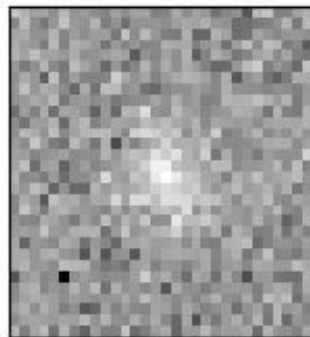
Processed



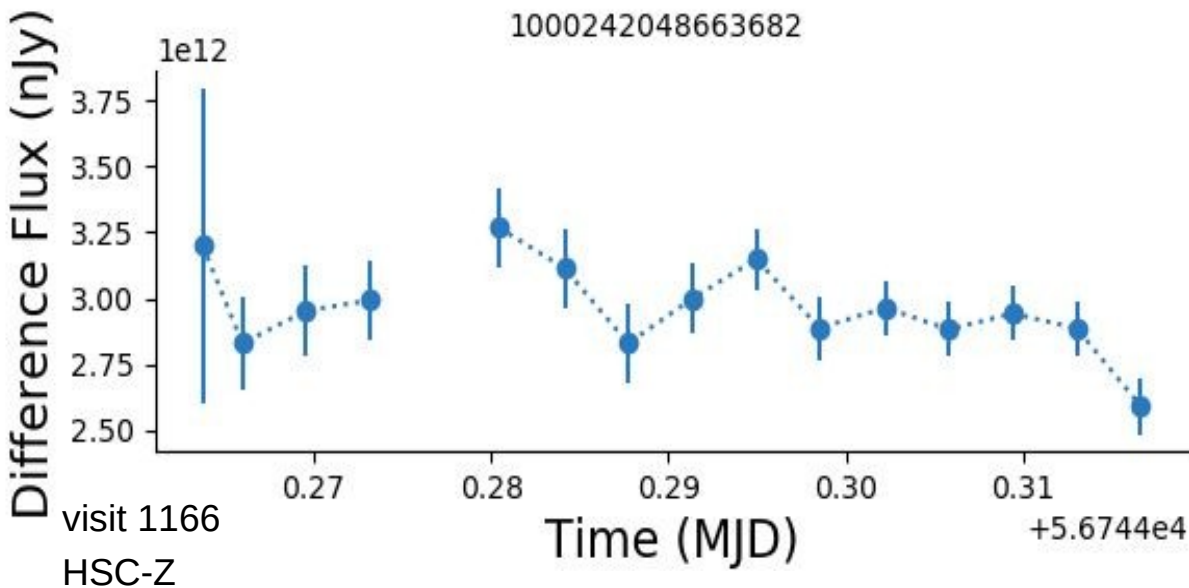
Template



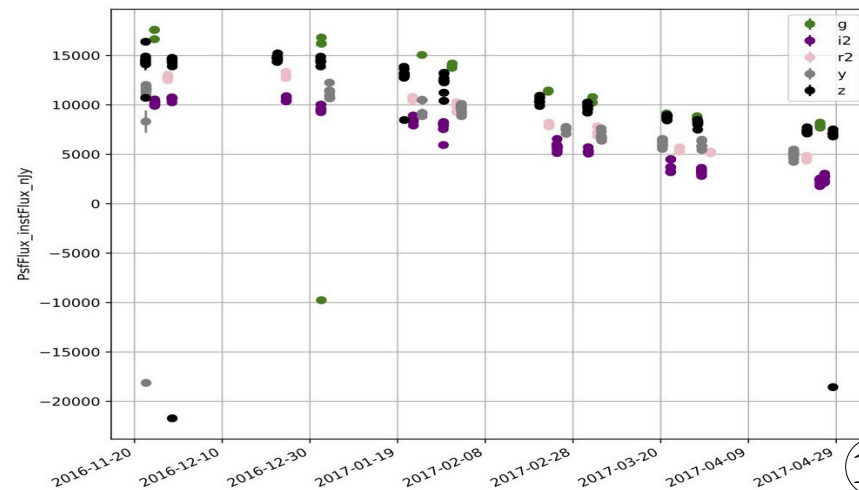
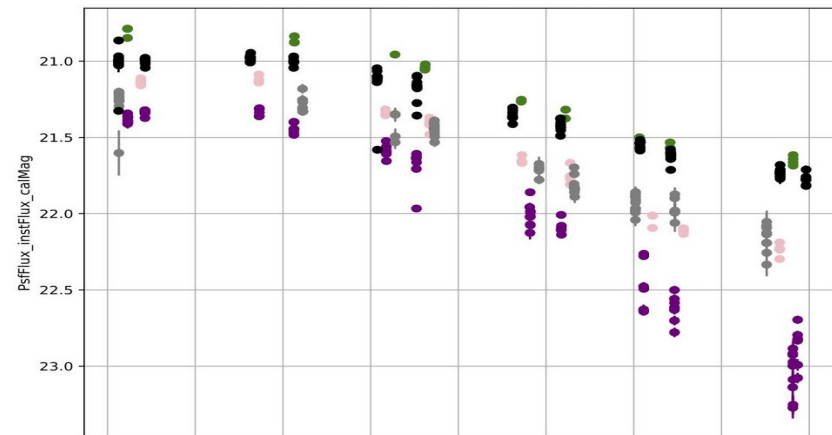
Difference



1000242048663682



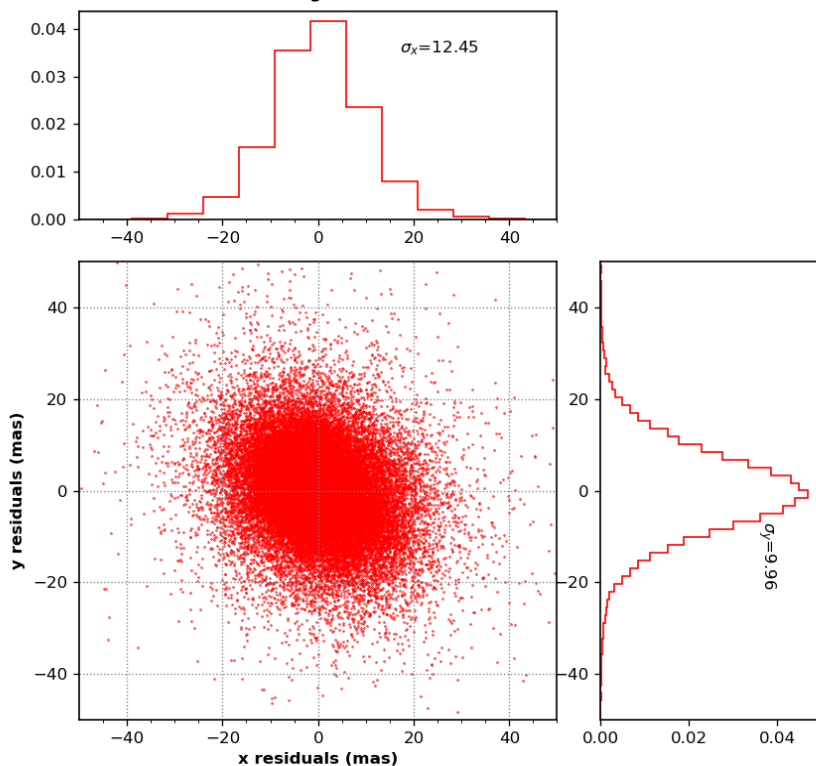
43158880816988319 around 2016-11-23



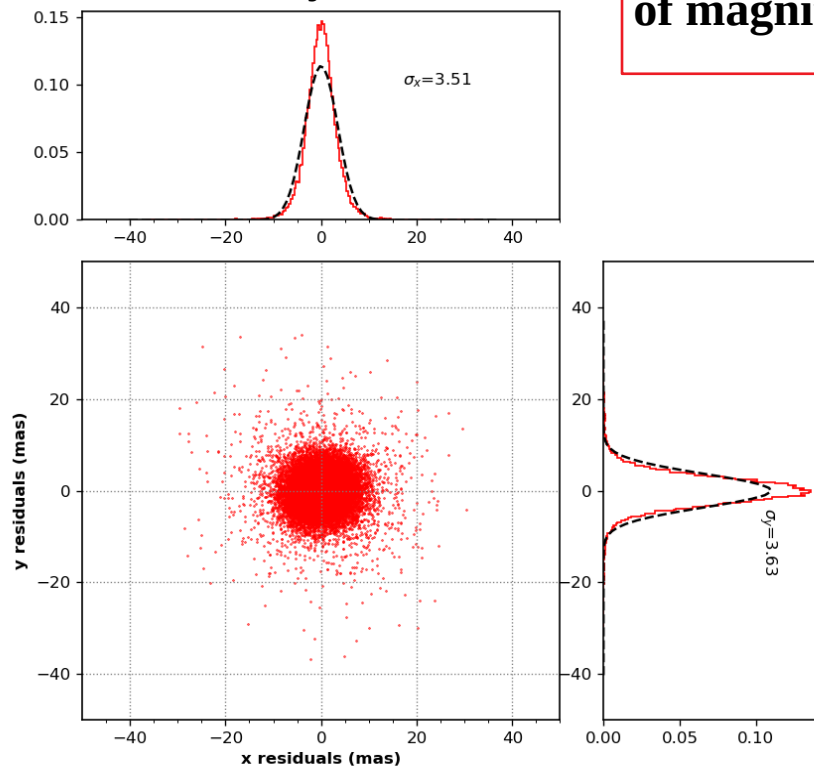
Jointcal: astrometric results

Residuals (mag<20), Night 56744, z band

before jointcal



with jointcal



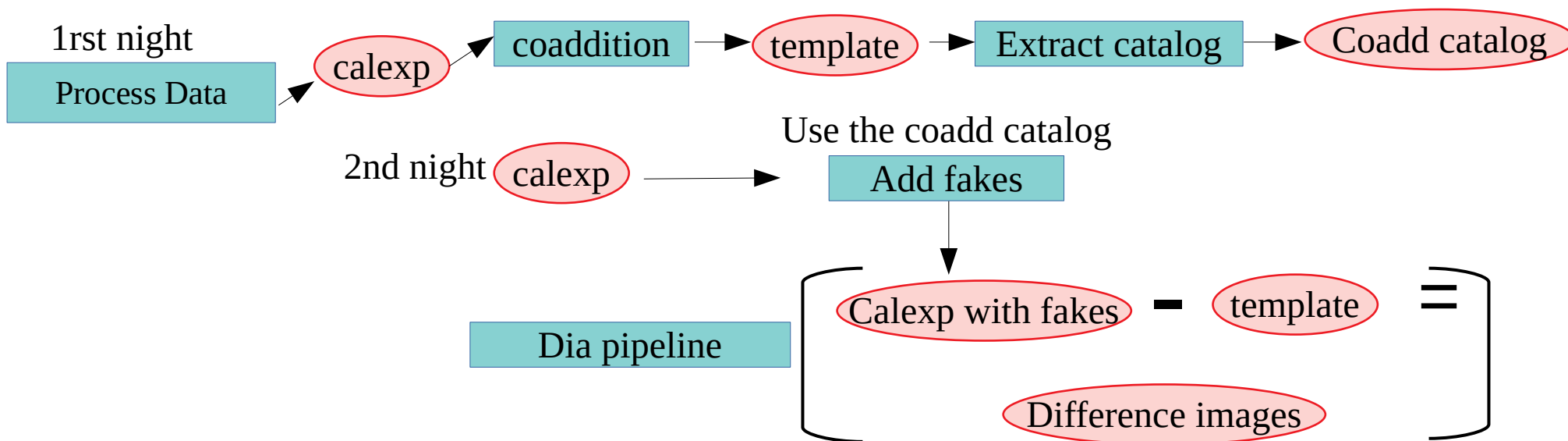
Jointcal reduces the error distribution by ~3 orders of magnitude

Dia-pipe versus ap-pipe

dia_pipe	ap_pipe
DESC product	“LSST pipeline” product
designed for “post-processing”	designed for “live” alerts streams
association updated after all visits	association updated visit per visit
Parquet files or Butler	database
Can’t use external templates?	Can use external templates
Not too hard to implement hacks	A bit hard to propagate hacks
Requires to load calexps for calibration	calibrated flux info in database: LC easy
Fast?	Slow?

Validation of the Dia pipeline

- from 2 nights, PDR1 UDEEP COSMOS field
 - Filter: Z filter
 - Process exposures for 2 nights (2014-03-28, 2015-01-16)
 - Coaddition over 1 night (2015-01-16) → produce template
 - Add fakes on 3 visits (night 2014-03-28, visits: 1166, 1168, 1170)



On-going work

Detect transients in the HSC images Study the SN Light Curves

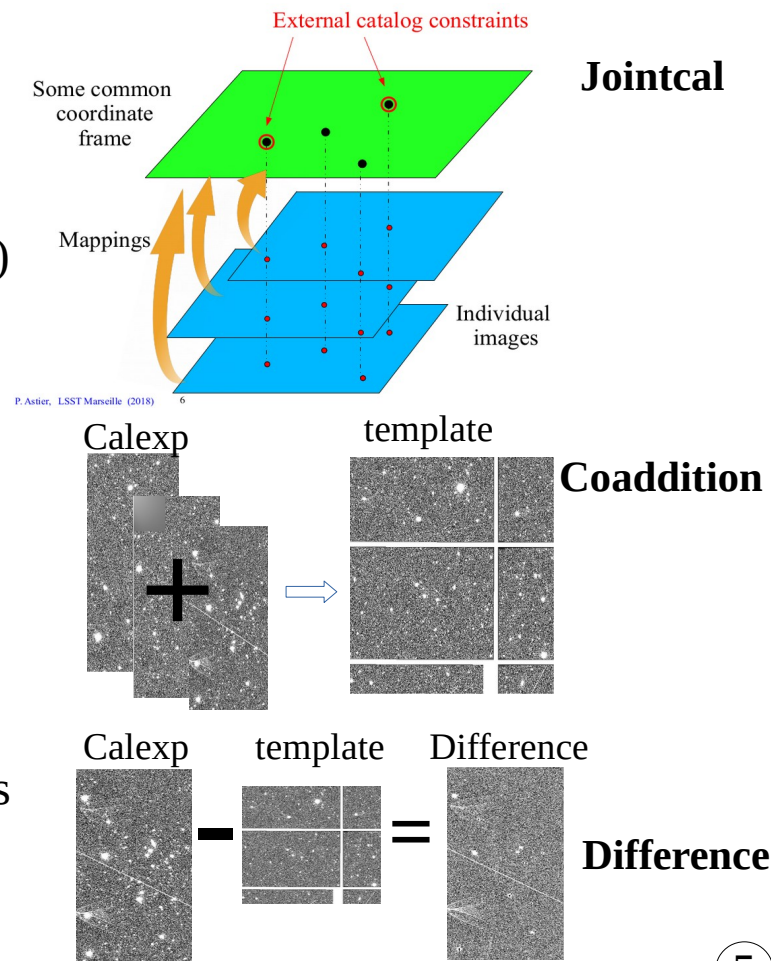
- Process HSC data, Cosmos_Ultradeep, 3 nights in R and Z (DR1)
- Use jointcal: photometric and astrometric calibrations
- Coaddition of the exposures (with and without jointcal)
- Difference images

Results !

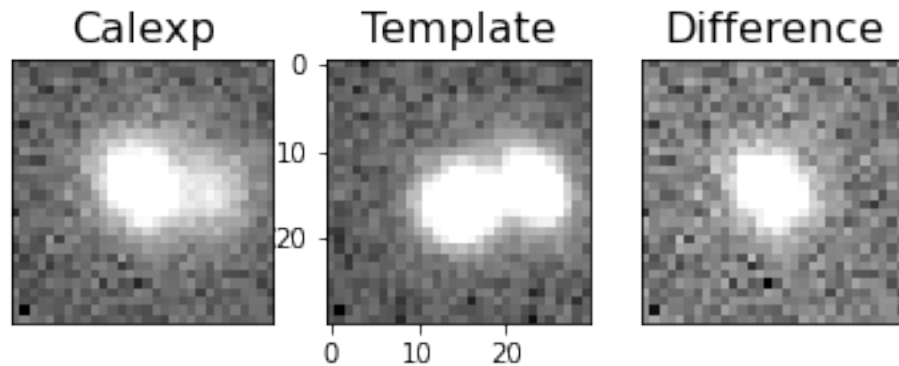
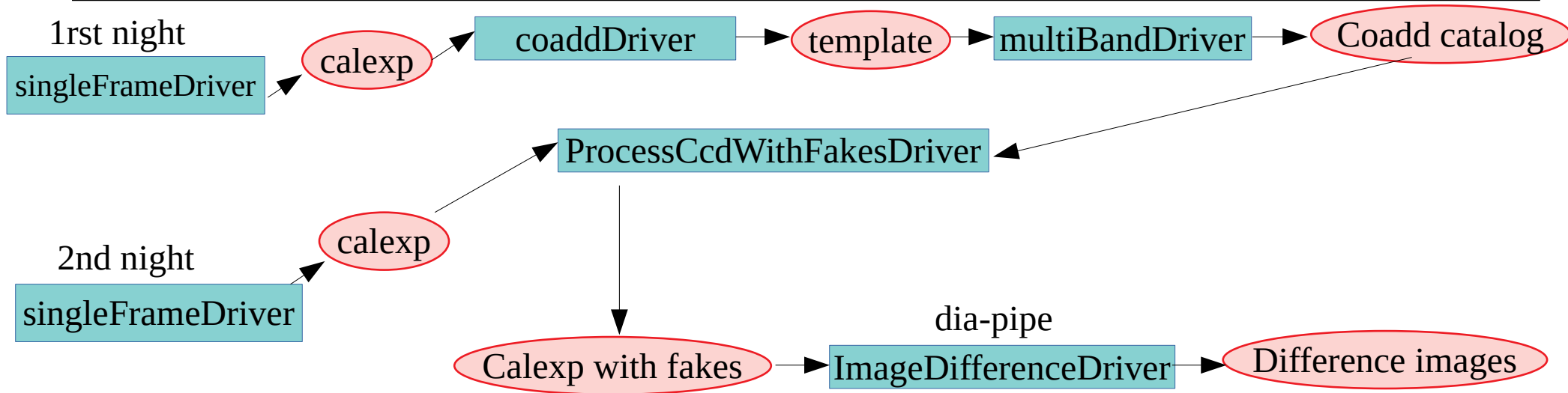
See [Control plots with Jointcal](#)

To-do list:

- Use jointcal with Gaia (as reference catalog)
- Waiting for DR2 data on CC
- Search the transients in the HSC Dia images, find their light curves
- Study the validation of photometry and detection efficiency of SN

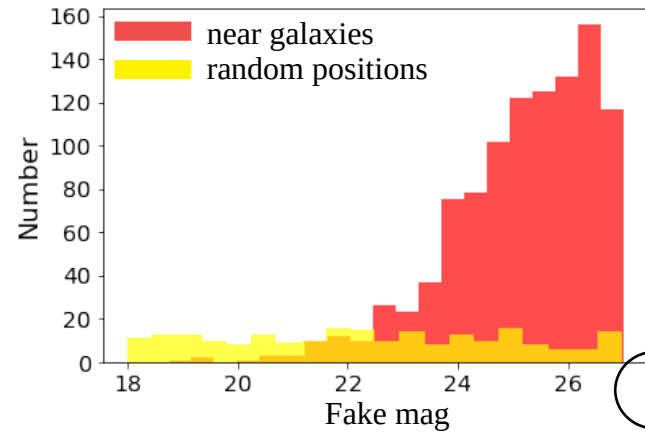
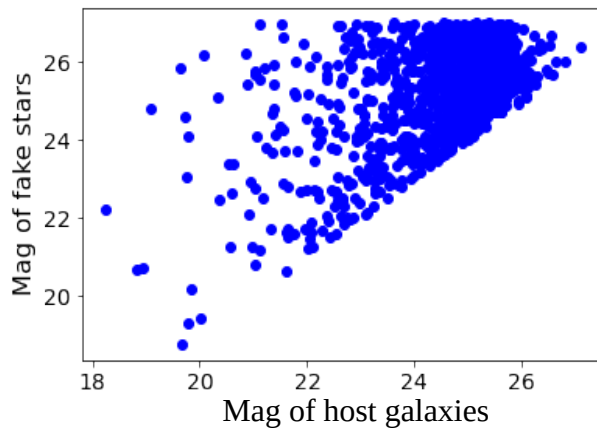
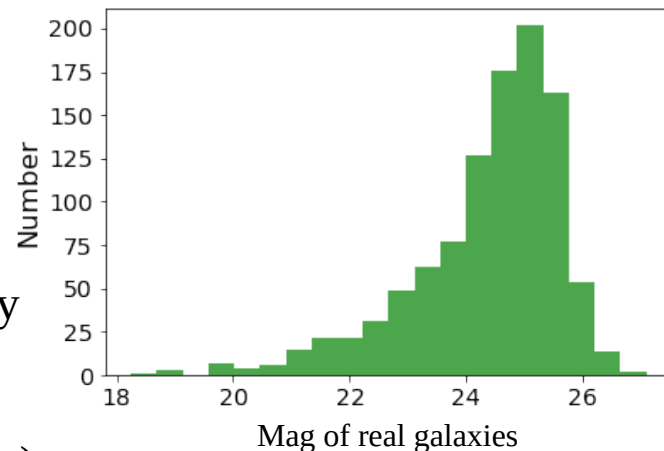


Pipeline



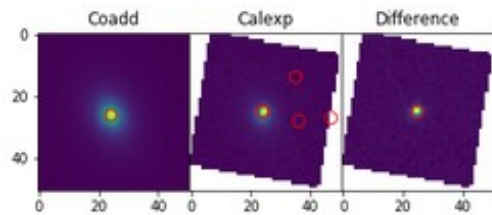
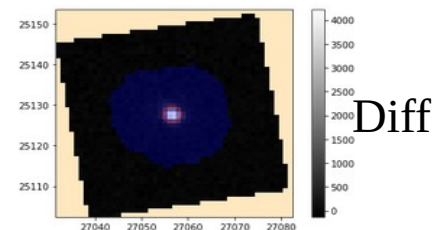
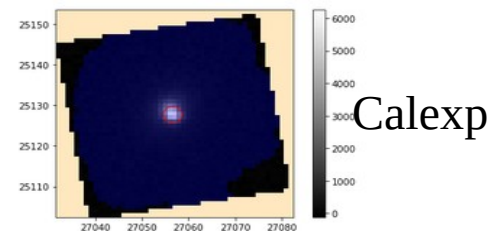
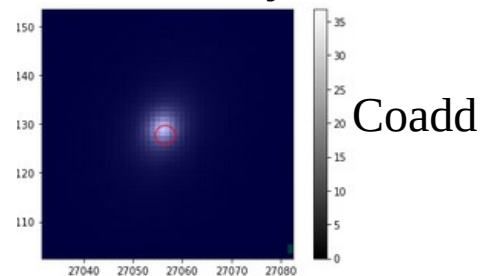
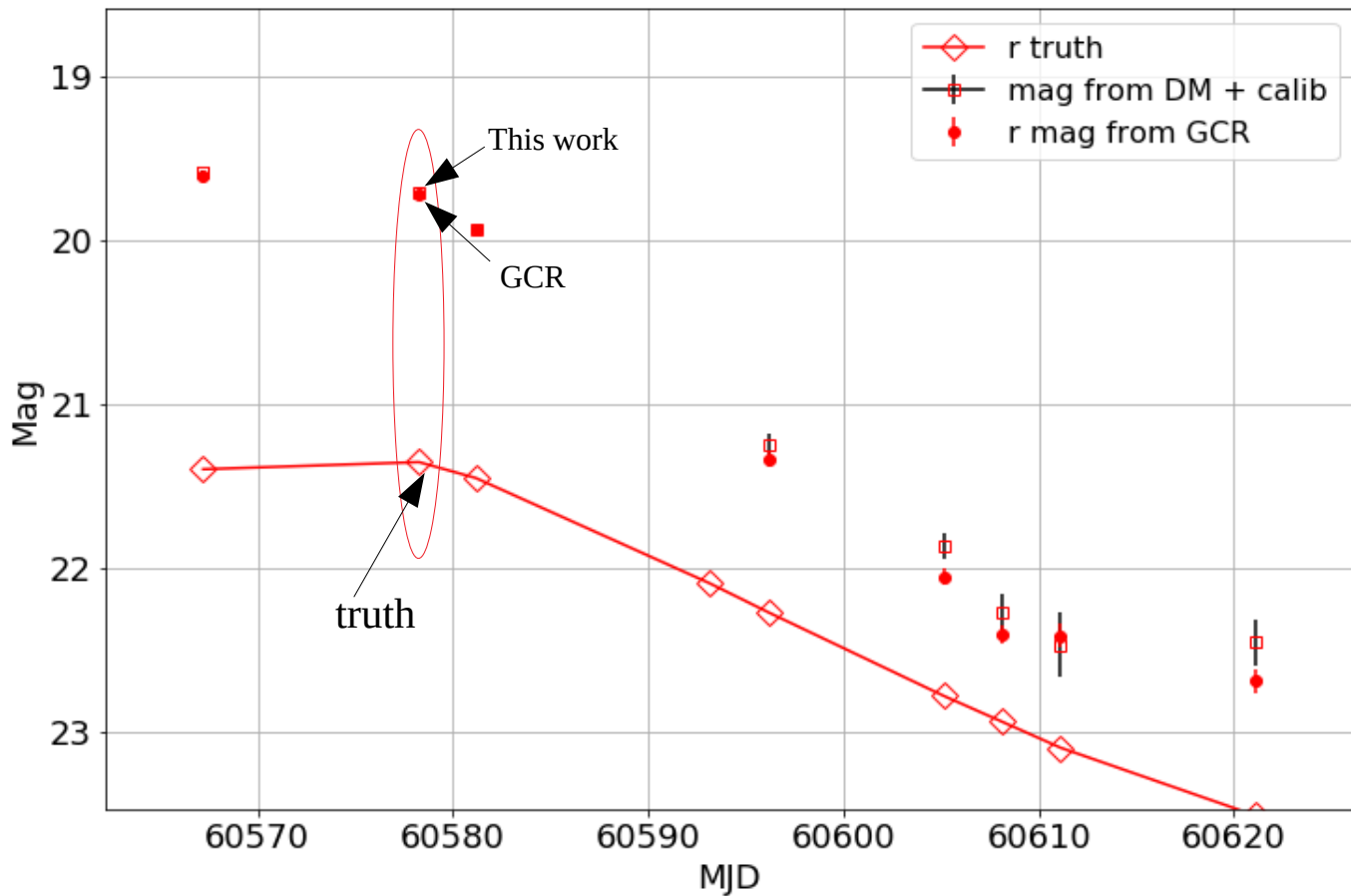
Fake injection

- Among the coadd catalog of an HSC sky-slice patch 5,5, tract 9813
 - ➔ Select galaxies (without deblending)
 - ➔ Excluded flagged sources ('base_PixelFlags_flag', 'base_PsfFlux_flag')
- Generate fakes associated to a galaxy (on 10% of randomly chosen galaxies)
- Gaussian distribution to choose random position according to galaxy radius
- Flat distribution to choose mag ([galaxy mag -1, 27])
- Generate 2% fakes with random positions (without galaxy association)
- Check the detection within magnitude intervals



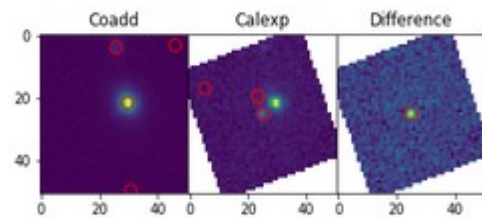
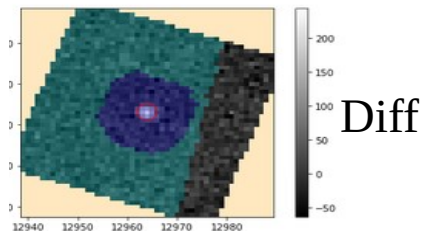
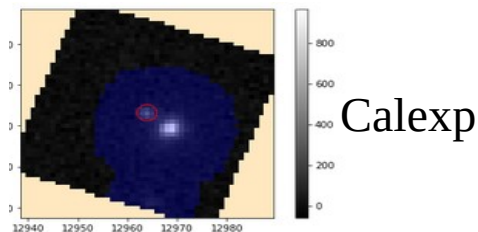
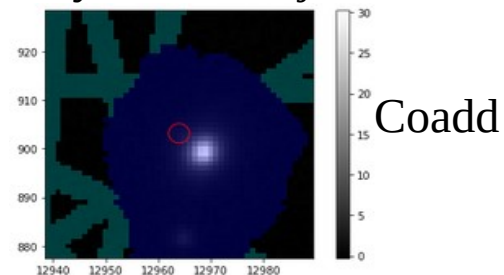
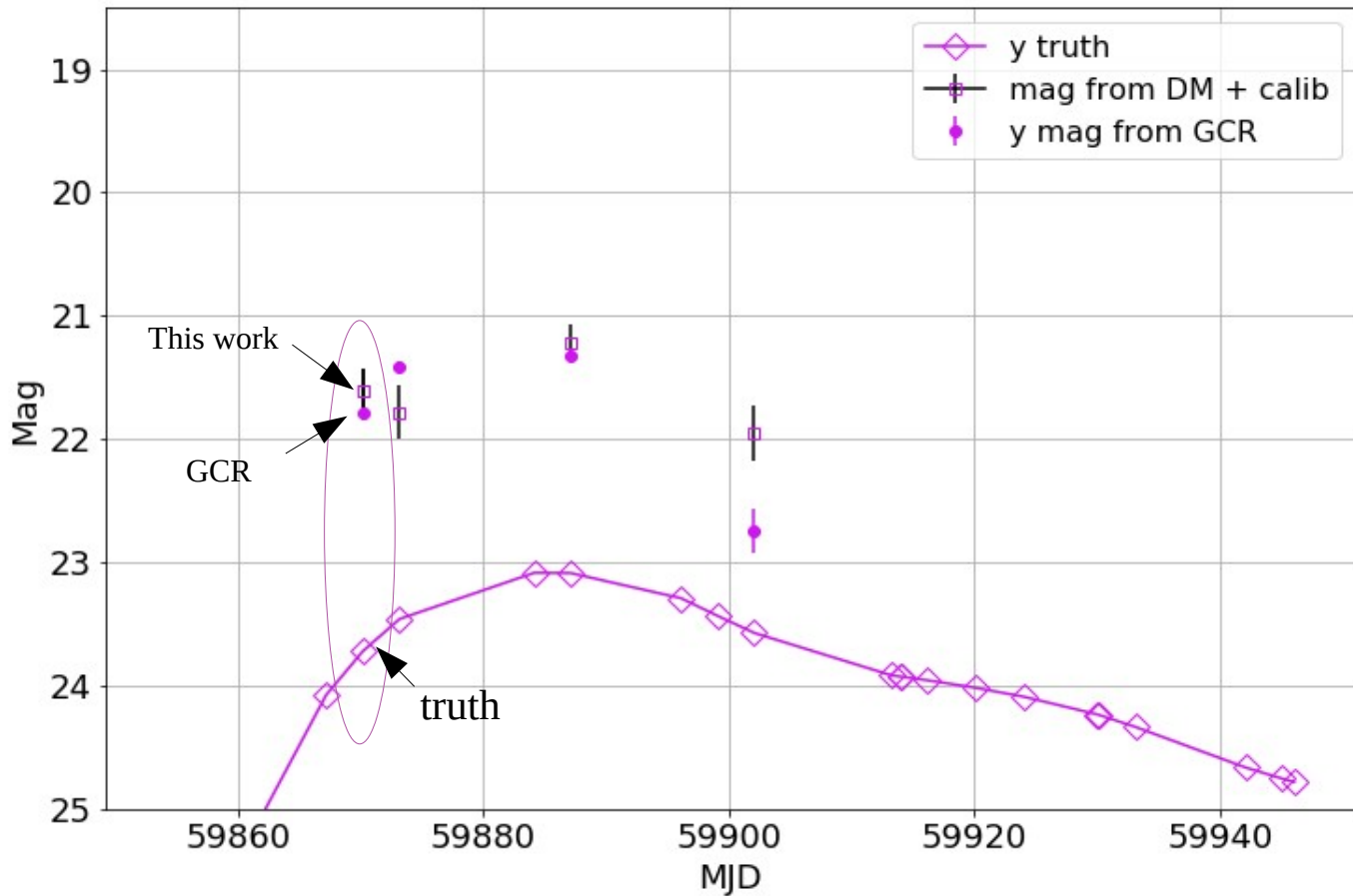
Using the ZP of each visit

for visit: 671659 filter: r ccd: 45 mjd: 60578



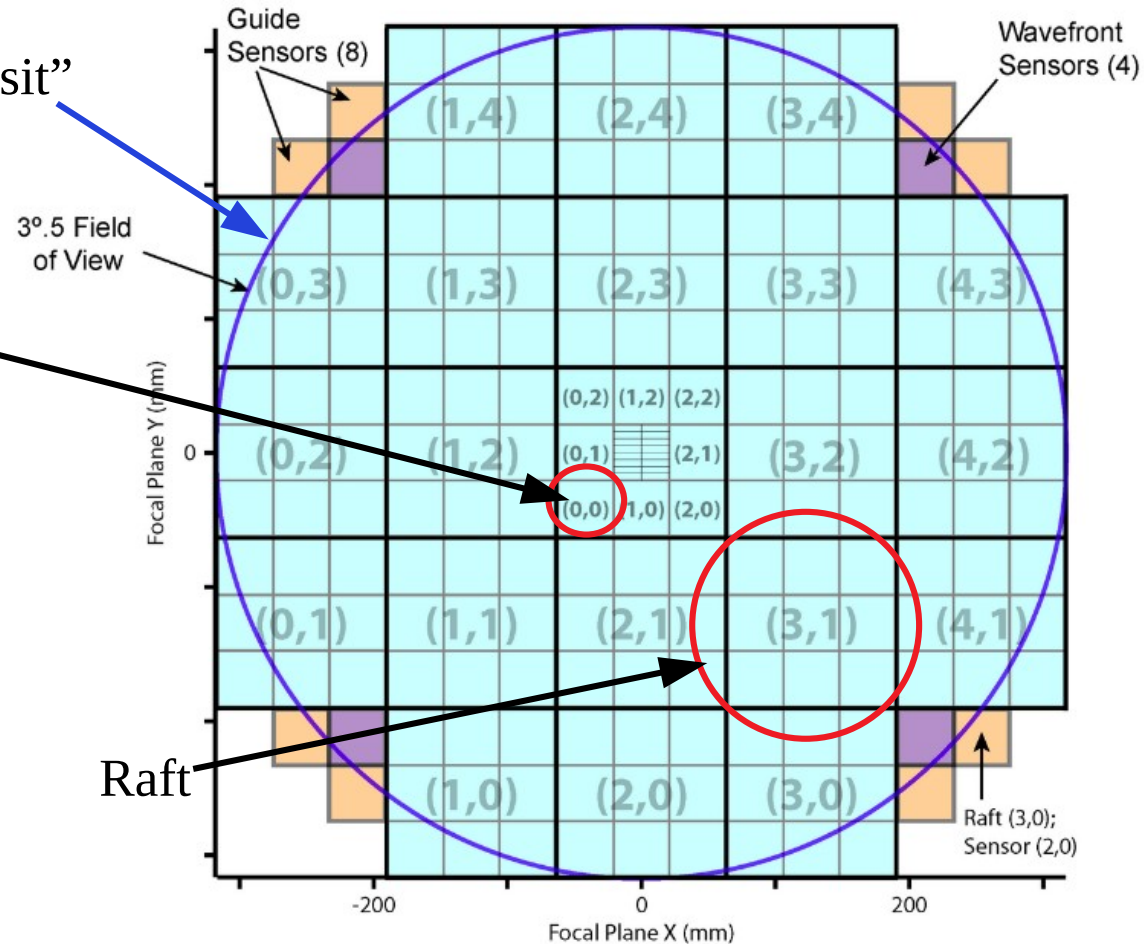
The truth object is 0.01arcsec away

for visit: 214391 filter: y ccd: 84 mjd: 59870



The LSST Focal Plane

- LSST observations are taken as a pair of 15-second “snaps” which constitute a “visit”
- The image data for a visit is a set of 189 “CCD” or “sensor” images.



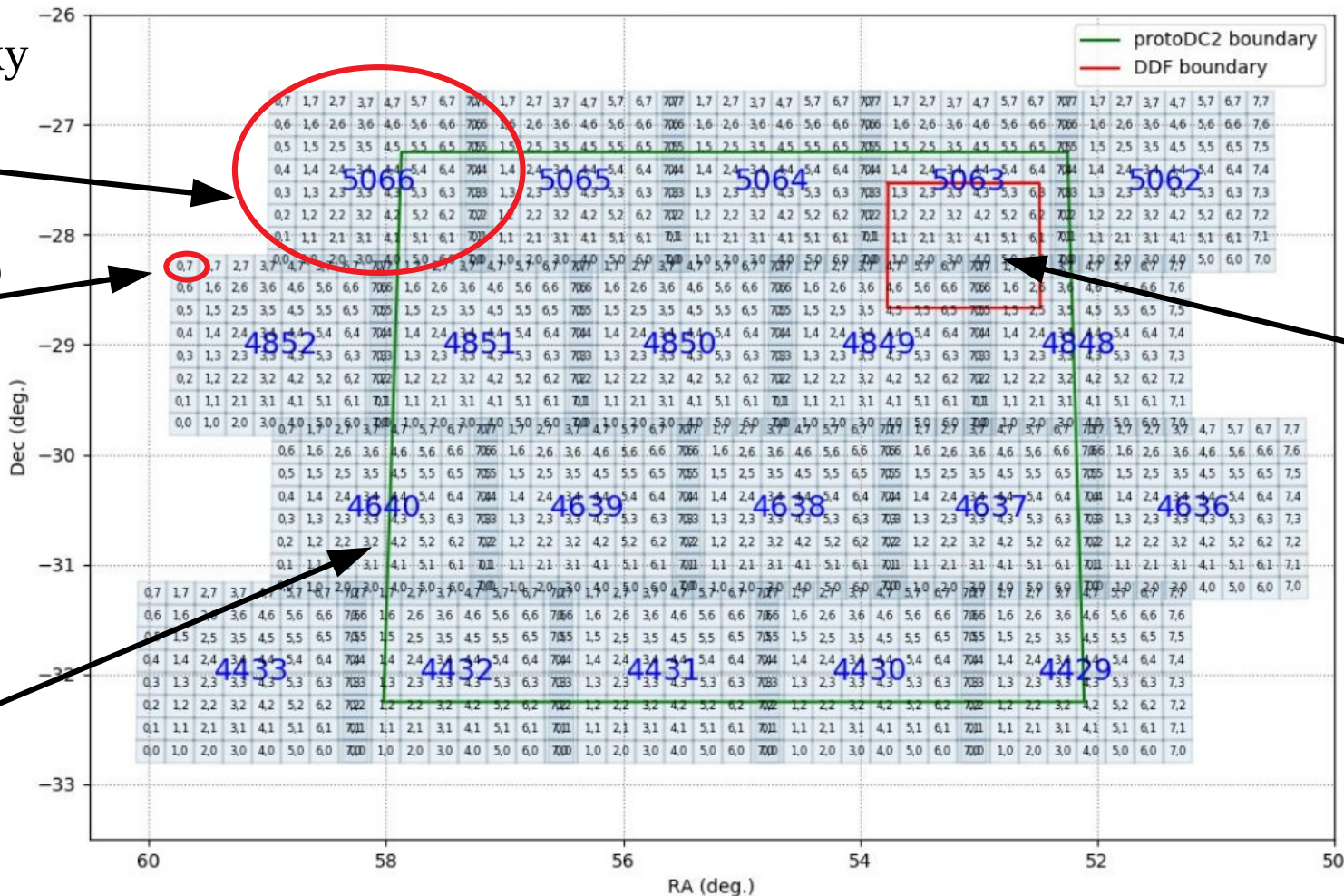
The LSST simulated sky region

The regions on the sky are defined with the concept: “**tracts**”

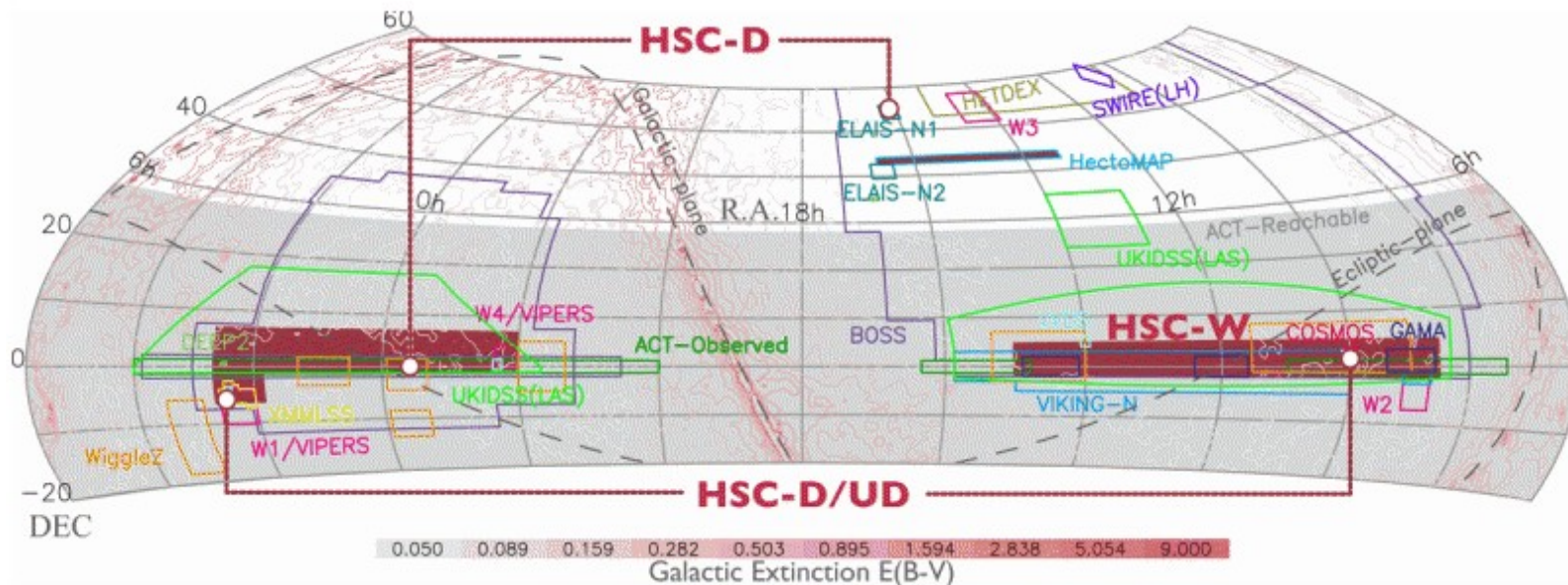
They are divided into “**patches**” which share the tract coordinate system

WFD

DDF



HSC survey field



HSC Survey Fields selected based on

- Overlap with SDSS regions, and other interesting datasets (CMB, NIR, BOSS, ACT, XMM, GAMA, VVDS, VIPERS)
- Low dust extinction
- Spread in RA

DR1 released in Feb 2017

DR2 released in Jan 2020

DATA

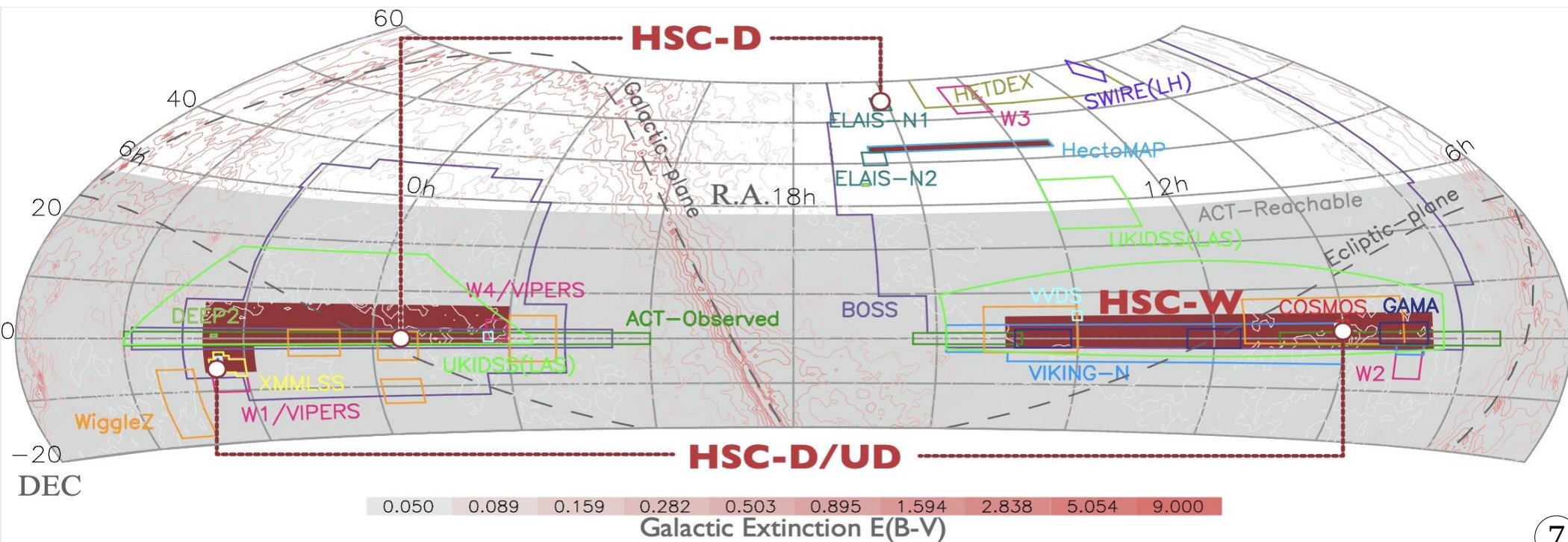
Release : HSC-SSP second public data release (**PDR1**)

Field : **COSMOS** (3-3.5deg² , $r \sim 27.7$)

Layer: **UDEEP**

Filters: R Z

- Process Udeep Cosmos data in 3 nights:
56744 (Z:16 exposures, R: 12 exposures)
57099 (Z:16 exposures)
57038 (R: 7 exposures)



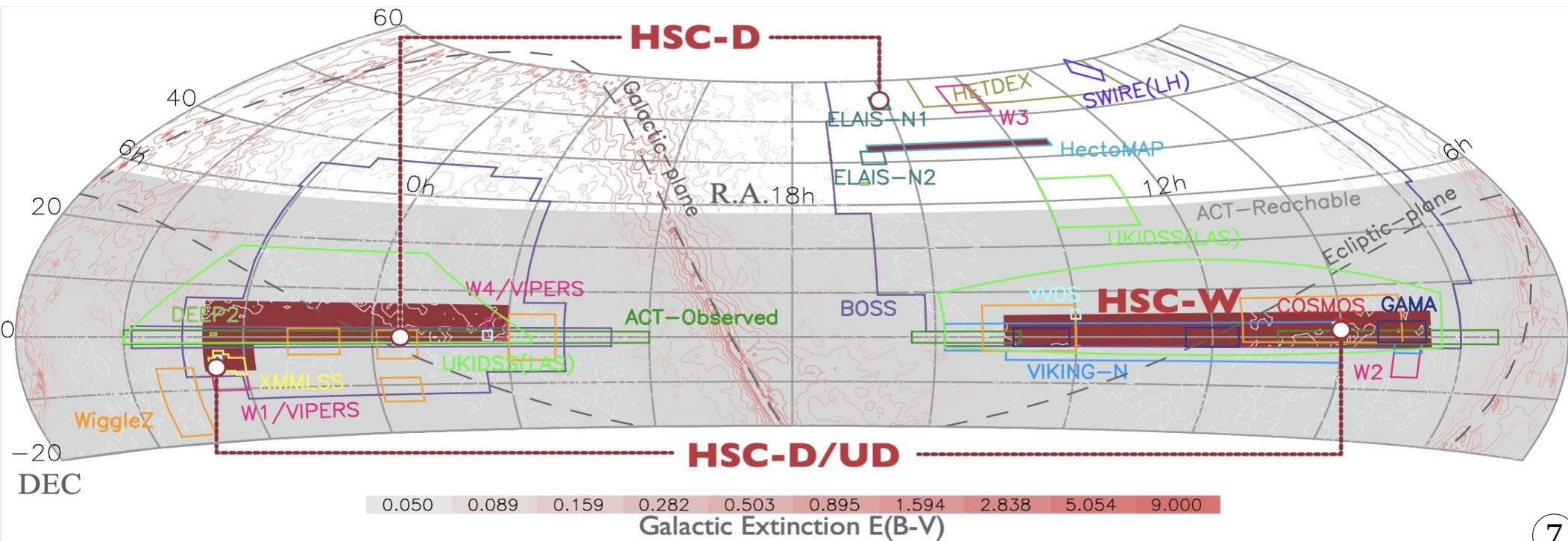
DATA

Release : HSC-SSP second public data release (**PDR2**)

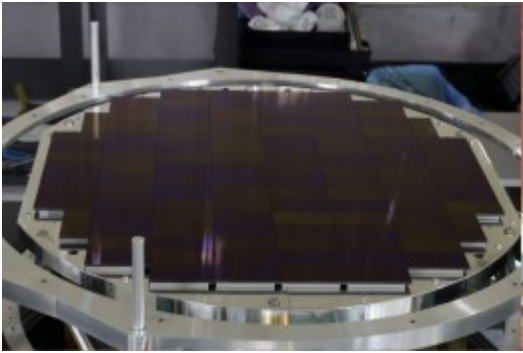
Field : **COSMOS** (3-3.5deg2 , r~27.7)

Layer: **UDEEP**

Filters: grizy (R Z)



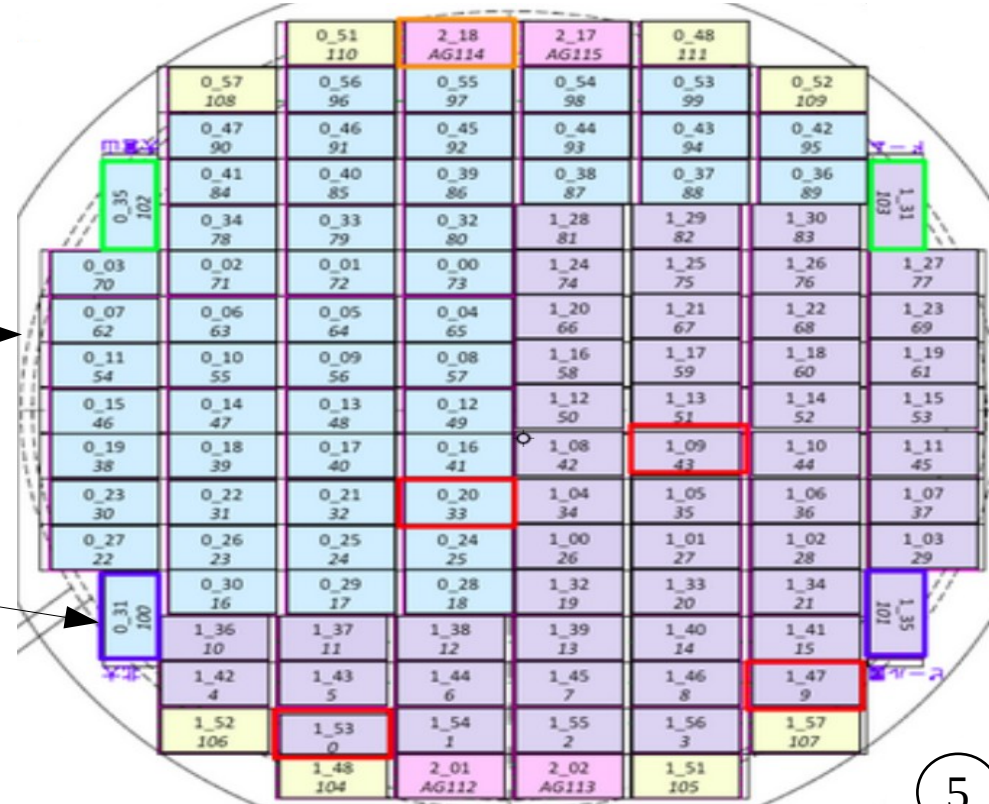
The HSC focal plane



The focal plane is filled with 116 CCDs:
104 for science, 4 CCDs for auto guider,
and 8 CCDs for auto focus

Camera observations called: “visits”

The image data for a visit is a set
of 116 “CCD” or “sensor” images.



HSC characteristics

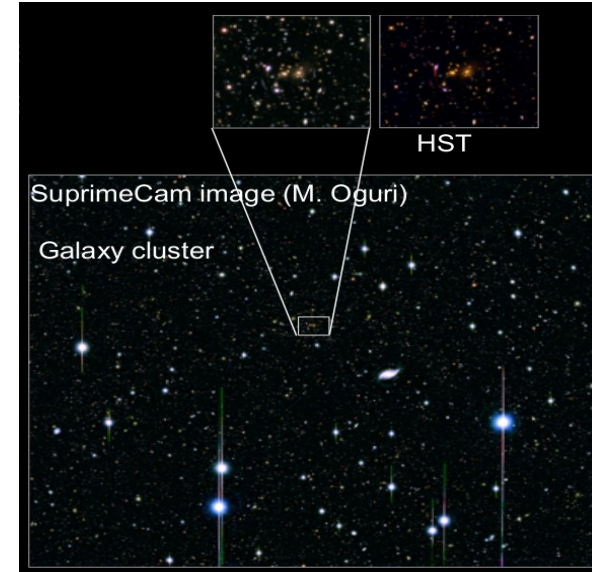


Fast, Wide & Deep

300 nights over 5-6 years (started in March 2014)

- grizY+4 narrow bands
- Wide FoV: $\sim 1.5^\circ$ diameter
- 3 layers:
 - Wide (1400deg^2 , $r \sim 26$)
 - Deep (27deg^2 , $r \sim 27$)
 - Ultradeep ($3\text{-}3.5\text{deg}^2$, $r \sim 27.7$)
- Excellent Image quality: $\sim 0.6''$ seeing

UltraDeep	<i>g</i>	<i>r</i>	<i>i</i>	<i>z</i>	<i>y</i>
exposure (min)	70	70	130	130	210
seeing (arcsec)	0.74	0.62	0.64	0.59	0.74
depth (mag)	27.5	27.3	27.2	26.5	25.7

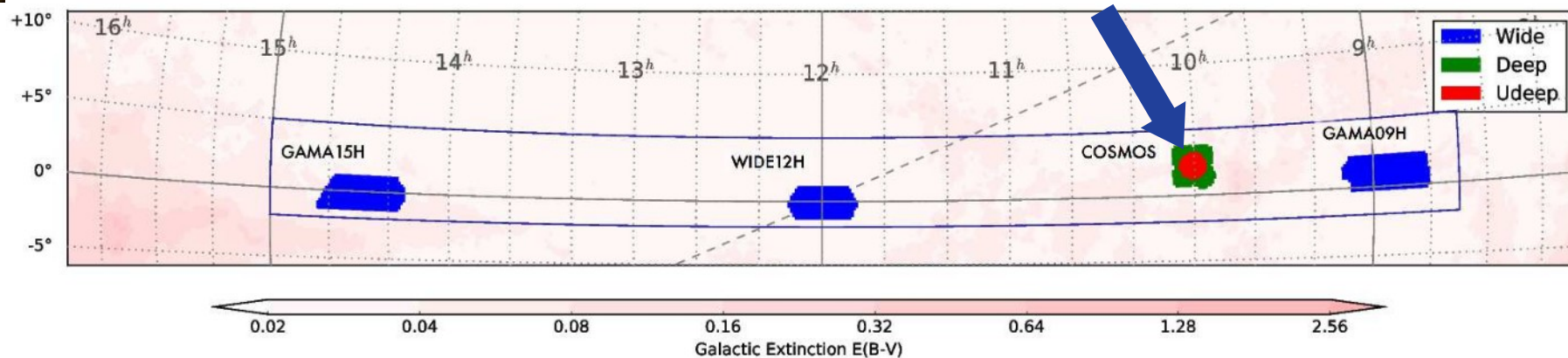
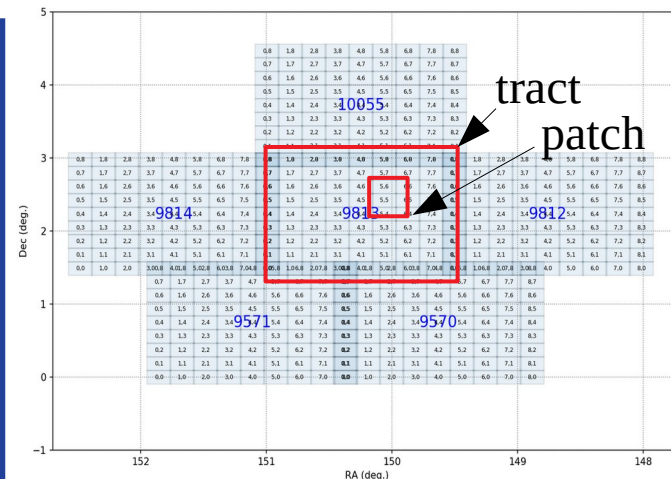


HSC characteristics



COSMOS: Cosmic Evolution Survey

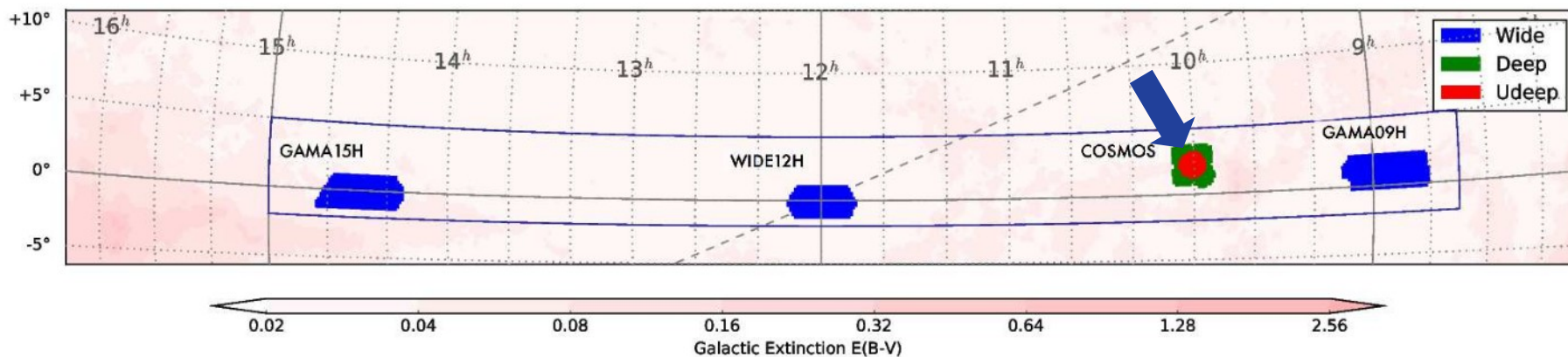
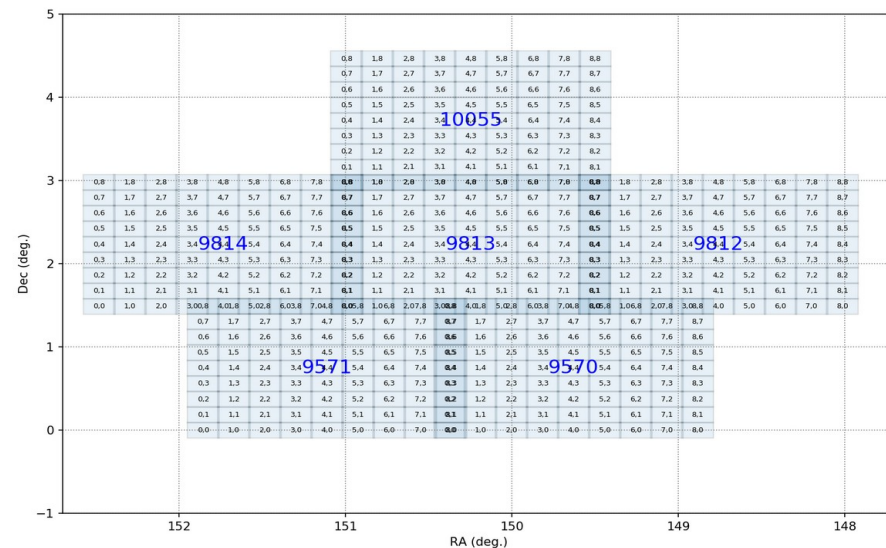
- Astronomical survey designed to probe the formation and evolution of galaxies as a function of redshift
- It covers a 2 deg² equatorial field
- ~ over 2 million galaxies are detected



HSC data

- Process Udeep Cosmos data in 3 nights (1st Release):
2014-03-28, 56744 (Z:16 exposures, R: 12 exposures)
2015-01-16, 57099 (Z:16 exposures)
2015-03-18, 57038 (R: 7 exposures)

UltraDeep	g	r	i	z	y
Exposure (min)	70	70	130	130	210
Seeing (arcsec)	0.74	0.62	0.64	0.59	0.74
Depth (mag)	27.5	27.3	27.2	26.5	25.7

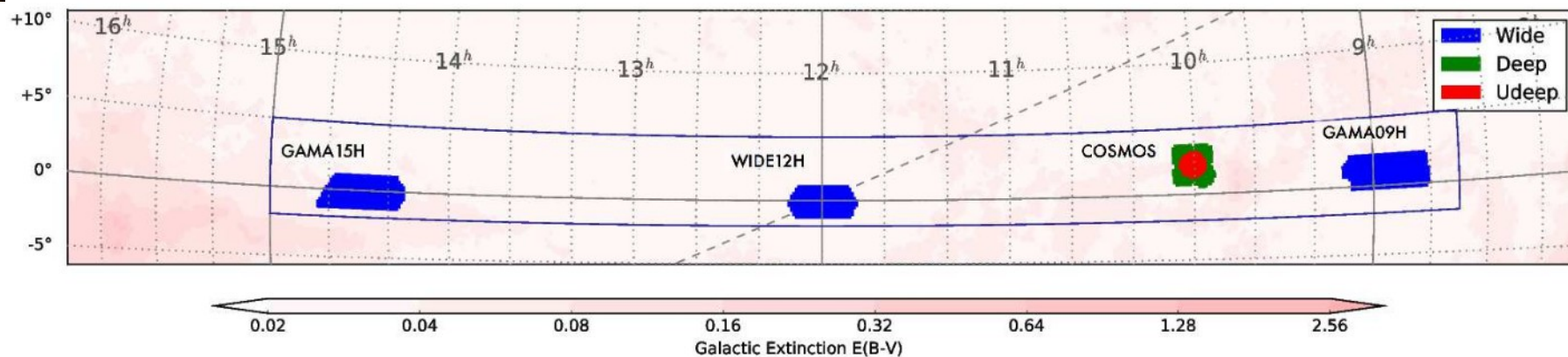
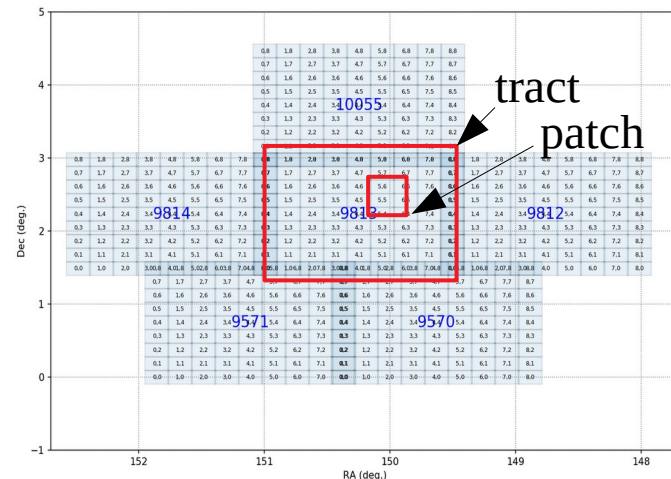


HSC characteristics

Fast, Wide & Deep

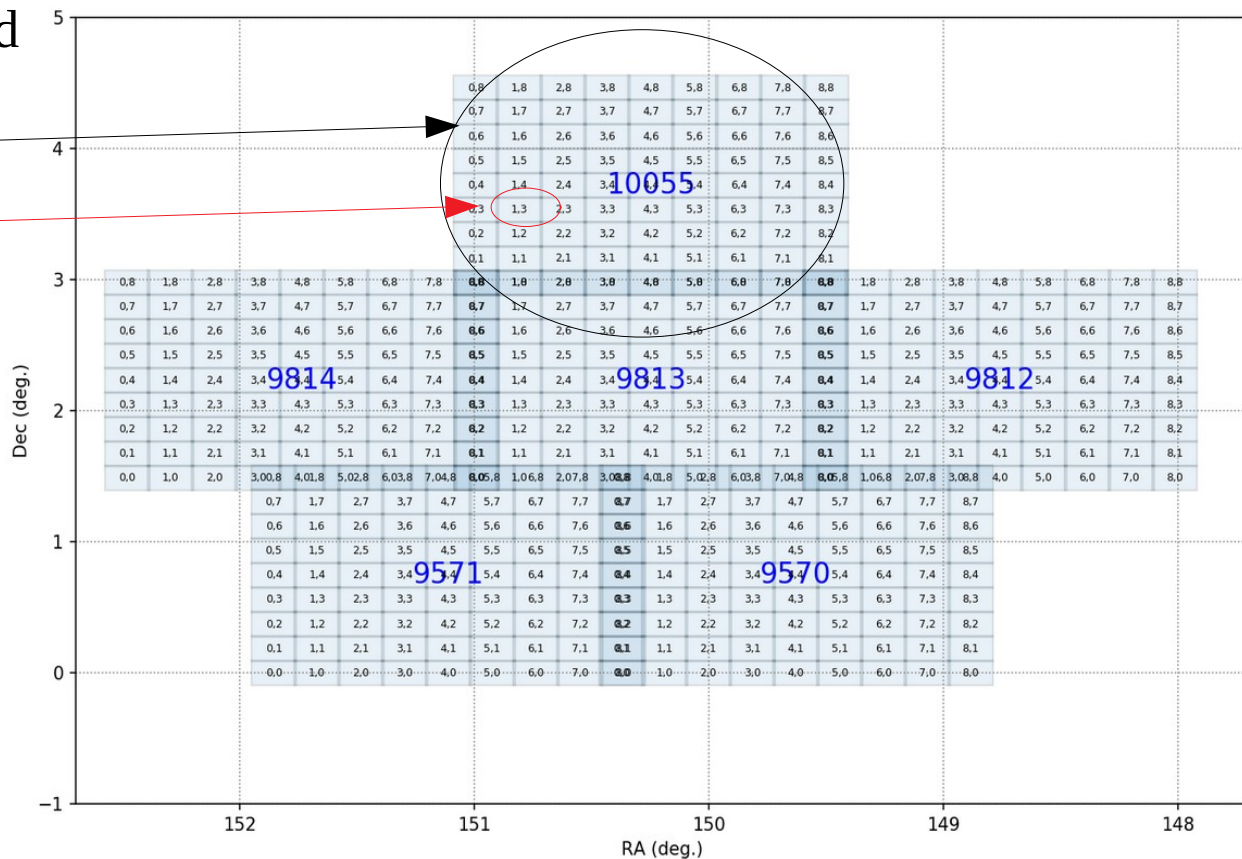
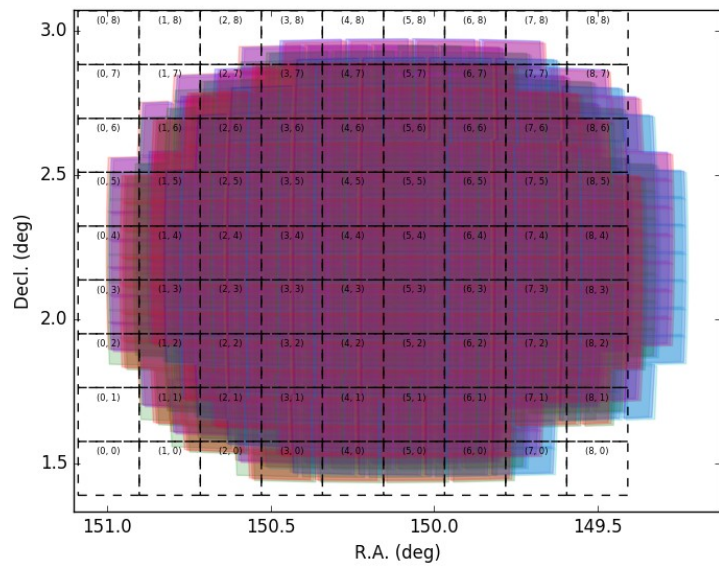
300 nights over 5-6 years (started in March 2014)

- grizY+4 narrow bands
- Wide FoV: $\sim 1.5^\circ$ diameter
- Excellent Image quality: $\sim 0.6''$ seeing
- 3 layers: - Wide (1400deg^2 , $r \sim 26$)
- Deep (27deg^2 , $r \sim 27$)
- Ultradeep ($3\text{-}3.5\text{deg}^2$, $r \sim 27.7$)



Sky region

- The regions on the sky are defined with the concept: “**tracts**”
- They are divided into “**patches**” which share the tract coordinate system



Fake injection

- Among the coadd catalog of an HSC-sky slice
 - Select galaxies (without deblending)
 - Excluded flagged sources
- Generate fakes associated to a galaxy (on 10% of randomly chosen galaxies)
- Gaussian distribution: choose a random position according to the galaxy radius
- Flat distribution to choose mag ([galaxy mag -1, 27])
- Generate 2% fakes with random positions (without galaxy association)
- Check the detection within the magnitude intervals

